

Resolution Improvement in GNSS Based Synthetic Aperture Radar

Using Signals from Multiple Satellites

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Dr. Najam Abbas Naqvi

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ABSTRACT

Direct Global Navigation Satellite System (GNSS) signals are used for positioning and navigation purposes while reflected GNSS signals are used in various remote sensing applications as they contain valuable information about the reflecting surface. The Global Positioning System (GPS) satellites, imaging hardware (GPS Rx) and its signal detection components (including antennas) forms a bistatic GPS based synthetic aperture radar (SAR) system. A single receiver located near or on the ground can collect direct and reflected signals coming from various GPS satellites and a set of different point target based scatter images of the interested area can be obtained by using SAR signal processing techniques.

This research endeavor describes a method to use reflected GPS L1 signals from one or more satellites to improve the resolution of simulated images. The entire GPS signal generation and image reconstruction process has been carried out through extensive simulations as diligently as possible within the limitations of the available computational resources.

There could be different imaging scenarios with different satellite geometries and all imaging scenarios were simulated to obtain point target based images from different satellite. These obtained images are later fused by a technique named as Stationary Wavelet Transform (SWT) image fusion in order to improve the resolution of simulated images.

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1 INTRODUCTION

The Global Positioning System (GPS) was developed by US Department of Defense (DOD) in order to give accurate estimation of position velocity and time to the US military [1]. GPS is currently fully operational and fits the criteria specified in the 1960s for an optimal positioning system. All users with appropriate equipment can obtain exact position, velocity, and time from the system. The constellation of GPS satellites has 24 orbiting satellites in 6 planes each having 4 satellites. GPS is able to provide its services to endless number of consumers. The Services for both civil and military users are provided by GPS. Standard positioning services (SPS) and precise positioning services (PPS) are two services given separately. The SPS is for civilian usage and the PPS is only for U.S. troops [2]. Reflected GPS signals also known as multipath signals are one of the frequently occurring error sources but can also be exploited for different remote sensing purposes since they carry vital information about the reflecting surface [3].

Many research publications have examined the possibility of employing reflected Global Navigation Satellite Systems (GNSS) signals as a remote sensing option. Many research initiatives are underway and published literature cover data collecting using different remote sensing technologies to monitor water surface and storage levels [4]. To detect soil moisture and effective height changes [5] for oil slick detection, ocean scatterometry, ice monitoring and soil moisture retrieval [6].

The study describes simulations of a SAR imagery system that uses GPS reflected signals from an object on the surface of the Earth to build an image of the interested area. GPS satellite, hardware for imaging and signal detecting components constitute a bistatic SAR system. As the GPS satellites are in constant motion and the received signal will be a

frequency modulated (FM) signal, which is to a continual change in Doppler shift and is also known as a chirp signal.

A chirp signal shows a narrow pulse effect with its auto-correlation function. The fundamental concept describes that resolution of the radar range is an inverse of signal bandwidth. They are used for obtaining a broadband with a reasonably large width of time. The chirp signal is a linear FM pulse and may be shown in linear frequency growth with time with the help of the following equation:

$$f(t) = \cos\left(\omega_0 t + \pi \frac{F}{T} t^2\right) \text{ for } -\frac{T}{2} < t < \frac{T}{2} \quad (1.1)$$

Where $F = f_2 - f_1$ is the sweeping bandwidth, ' f_1 ' and ' f_2 ' correspondingly are the first and last instant frequencies whereas ' T ' represents the time duration. As the FM GPS signal has superior pulse compression features, the scope and resolution of the SAR image system can therefore be achieved. A matching filter is used to improve the system resolution in the received signal processing system, as the matched filter is ideal for signal detection in the white noise environment [7].

Each signal that is reflected has a distinctive chirp that can be recognized and coherent to the direct signal, if the satellite's geometry in terms of signal detection and visibility is optimal. The targeted place will have a good correlation value and the area of interest can be determined. Thoroughly simulations have shown that a bistatic SAR system is able to produce images of the area with reasonable quality and with cheaper cost and their resolution can be improved by applying a suitable image fusion method.

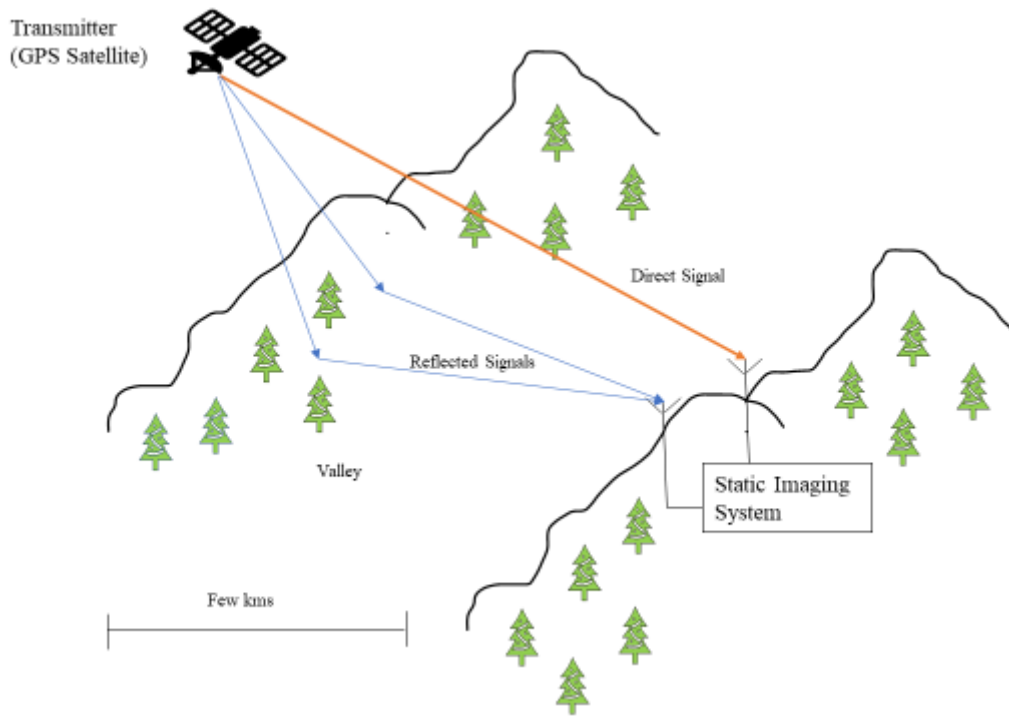


Figure 1. 1 Imaging scenario for static imaging system

An image of the field of interest may be generated by a static receiver if the sample is done for a longer period of time; this kind of case is shown in figure (1.1). The geometry shift is provided by the moving GPS satellite with the static receiver, although a change in geometry is rather slow and leads to very long data acquisition time in the order of hours.

The different configurations have been simulated, discussed and analyzed in a detailed manner during current research effort. Various research papers have highlighted the prospect of adopting this approach for imaging applications (some of them are discussed in literature review chapter), but there is little information regarding improvement in resolution of the obtained images. In such a context, simulation efforts to obtain images of the interested area and improving their resolution are a novel achievement.

1.1 OBJECTIVE

At the time of initiation of research, it was decided to study and simulate static imaging arrangements and scenarios and improve the resolution of obtained point target based scatters images in a suitable software environment using Matlab. The goal is to develop an appropriate image reconstruction algorithm, the noise and error source simulation, the acquisition of the GPS signal and the generation of images for different satellite geometries in the area of interest and to develop an image fusion algorithm that combines images to improve their resolution.

The aim of the thesis is to determine how an effective reconstruction code, extensive simulations and image fusion techniques have achieved this objective. This thesis illustrates how the correlation of direct and reflected GPS signals is attempted to generate point scattered images with different satellite geometries and based on SAR techniques and combining them by using Stationary Wavelet Transform (SWT) technique.

1.2 ADVANTAGES OF THE GPS BASED IMAGING SYSTEM

Below are the few advantages of the GPS based imaging system:

- a) The GPS-based object detection and imagery attracts any user who has been sustained on the pricey GPS navigation infrastructure and no specific transmitter is necessary.
- b) Simultaneous imaging chances exist for every GPS satellite in view. For direct and reflected GPS transmissions, the GPS satellites have an optimal signal strength and visibility.
- c) Azimuth resolution can be improved by obtaining and overlaying the findings from two or more distinct GPS satellites.

- d) The system offers low power consumption as compared to huge dedicated transmitters.
- e) GPS operates continuously and covers the whole earth surface. This constellation provides users with 5 to 8 satellites visible on the globe from anywhere [8].
- f) Being a cost effective system is another advantage. The imaging equipment can be designed at a fraction of the price of conventional radars, space systems and other sensors, comparable in size in complexity to a laptop computer.

2 THEORY OVERVIEW

2.1 The Global Positioning System (GPS)

2.1.1 A General Description of GPS

The GPS was developed as a Military Navigation system by US military to guide targeted missiles, aircrafts and for other military applications. The program was initially started in 1960s with an aim to have following attributes: global coverage, high accuracy, continuous weather operations and ability to serve high dynamic platforms. In 1973, after approval of Department of Defense (DOD), the first satellite was launched in 1978 and The system had been deemed fully functional having 24 satellites in its constellation in 1995 with approximated development cost of about \$10 billion [1] [2].

2.1.2 Architecture of the GPS System

The GPS comprises of three segments.

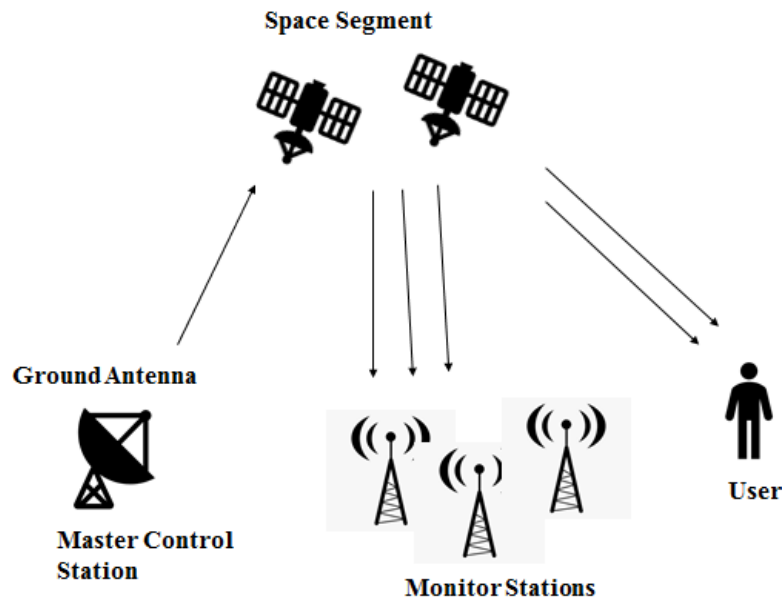


Figure 2. 1 Segments representation of GPS

One is space segment, comprises of 24 GPS satellites orbiting around earth at an altitude of 20,200 km from Earth's surface in six planes with four satellites in each plane. One orbit takes around 11 hours and 58 minutes to complete. The system is configured to show at least five satellites from Earth's surface at any time without any obstruction [2]. The GPS satellite constellation is made up of both ancient and new satellites. As of February 2020, 31 operational GPS satellites are in constellation [9].

A Master Control Station (MCS), Monitor Stations (MS), and Ground Antennas (GA) in various sites across the world make up the control segment. The purpose of MCS is to manage satellite operations, to maintain satellites a proper orbital position and monitor the health and status of satellite subsystems. The user segment is made up of GPS receivers, which are devices that are designed for receiving and processing of signals from GPS [2].

2.1.3 Principal of working

A GPS satellite consists of very high accuracy atomic clocks and codes are transmitted at a very precisely known time. In order to determine user's position, GPS make measurements by using the concept of time-of-arrival (TOA) ranging. Measurements are made to calculate the time taken by signal transmitted by a SV at a known location to reach a user's receiver. The position is calculated by measuring the propagation time of signals transmitted from multiple GPS satellites from known locations. The user's position is calculated by a simplest and more accurate method of location an unknown position, called trilateration [2].

In case we want to formulate the mathematics of satellite navigation problem, it is necessary to choose a reference coordinate system in which the states of both the satellite and the receiver can be represented. To measure and determine the orbits of GPS satellites, Earth Centered Inertial (ECI) coordinate system is used in which the origin is at the center of the mass of Earth and axes are pointing in fixed directions according to stars. However to measure the position of a GPS receiver, it is convenient to use Earth Centered Earth Fixed (ECEF) coordinate system that rotates with the Earth. In such system, it is easier to calculate the latitude, longitude and height parameters displayed by a GPS receiver. The x-y plane of ECEF coordinate system is coincident with the Earth's equatorial plane. It is necessary to have a physical model describing the earth to carry out the required transformations. A standard physical model of Earth that is used for GPS applications is World Geodetic System 1984 (WGS-84), which is a state of the art geodetic reference system and represents an ellipsoidal model of the Earth's shape.

2.1.4 Orbits

It is required to know the precise position of the GPS satellite in order to determine navigation parameters. This is why it's critical to comprehend GPS orbital

characterization. There are several methods of representing a satellite's orbital parameters. The GPS ephemeris data includes not only the six orbital parameters, but also the time when they are applicable and a description of how they vary over time. A GPS receiver can use this information to solve the navigation problem by computing the GPS satellite's corrected integral of motion [2].

Table 2. 1 Ephemeris data definitions

Symbol	Description
T_{oe}	Reference time of ephemeris
$\sqrt{a}$	Square root of semi-major axis
E	Eccentricity
I_o	Inclination angle (at time t_{oe})
Ω_o	Longitude of the ascending node (at weekly epoch)
Ω	Argument of perigee (at time t_{oe})
M_o	Mean anomaly (at time t_{oe})
di/dt	Rate of change of inclination angle
$\dot{\Omega}$	Rate of change of longitude of the ascending node
Δn	Mean motion correction
C_{uc}	Amplitude of cosine correction to argument of latitude
C_{us}	Amplitude of sine correction to argument of latitude
C_{rc}	Amplitude of cosine correction to orbital radius
C_{rs}	Amplitude of sine correction to orbital radius
C_{ic}	Amplitude of cosine correction to inclination angle
C_{is}	Amplitude of sine correction to inclination angle

The orbital elements used in the algorithms by which a GPS receiver calculates the position vector of a satellite (x_s, y_s, z_s) in the ECEF coordinate system are defined in Table (2.1), these parameters will be discussed further in the simulation chapter. Three elements describe the shape of satellite orbits, while three elements describe the orientation of the orbits in ECEF coordinate system. The orbits are circular with eccentricities of no larger than 0.02 and semi major axes of approximately 26,560 km's in case of GPS satellites. The remaining parameters vary between satellites so that the constellation provides uniform coverage of the whole Earth. The almanac and ephemeris data transmitted by the GPS satellite includes osculating Keplerian orbital elements that are augmented by "correction parameters" allow the user to estimate the Keplerian elements accurately during the periods of time when satellites ephemeris message is updated. It is important to know the Earth's rotation rate in to carry out necessary calculations. The rotation rate of earth is $\Omega_e = 7.292115 \times 10^{-5}$ rad/sec as described by WGS-84.

2.1.5 Services

The GPS satellites transmit two types of signals, one is Precision or P-code and other is Coarse Acquisition or C/A code. The P-code is referred to as Precise Positioning Service (PPS) designed only for authorized military users. It transmits signals on both L1 and L2 frequencies and provides maximum accuracy to authorized users. The P-code uses an encryption mechanism called Anti-Spoofing (AS) to prevent unwanted users from obtaining the P-code. Standard Positioning Service (SPS) is the C/A code that is designed for non-military usage. It is less accurate and transmits signals only on L1 frequency. This code is easier to acquire and the US military can reduce its accuracy by employing a method known as Selective Availability (SA). SA regulates the level of correctness offered to all C/A code users [10].

The chipping rate is the major distinction between C/A and P-code. A PRN Gold code of 1023 chips modulates the C/A code at 1.023 MHz chip rate, resulting in a bandwidth of 2.046 MHz and a repetition time of 1 millisecond. The Modulo-2 addition of two code sequences of 15,345,000 chips and 15,345,037 chips at a chipping rate of 10.23 MHz results in a bandwidth of 20.46 MHz for the P-code. [11].

Ionospheric delay error is the most common source of error in GPS positioning and navigation. When GPS signal travels through ionosphere, It has an additional time delay, this is a result of the ionosphere's Total Electron Content (TEC). In P-code this error can be mitigated by using dual frequency GPS receiver (L1 at 1575.42 MHz and L2 at 1227.60 MHz) through a linear combination of L1 and L2 observations [12]. That's why when a single GPS receiver is used for navigation purposes P-code performs more accurate than C/A code.

Differential GPS (DGPS) is another technology used to improve the accuracy of GPS for SPS users. In this strategy, a receiver is placed on a known-location reference station, and the GPS signal is observed in real time using the mobile station receivers. The GPS position signal is continuously received at the base station and then compared with its accurate data to obtain various correction values. The correction values are transmitted to the mobile station and finally the observation value of mobile station is corrected by itself to obtain accurate positioning data, which is the basic principle of differential GPS technology [13].

Differential GPS is made to improve accuracy of navigation in a local area. By applying the above mentioned corrections a user can get the accuracy of about 1-3 m. However, this is effective only in area of 100 km's of the reference station. Therefore, another technique Wide Area Differential GPS (WADGPS) is becoming popular which works in

a wider area while reducing the number of reference stations it gives an accuracy of 1.5-4 m for dual frequency user [14].

2.1.6 A Brief Introduction to C/A Code

The C/A code is made up of a 1023-bit pseudorandom noise (PRN) sequence with a clock rate of 1.023 MHz and a one-millisecond repetition duration. Each GPS satellite is given a unique PRN, which is chosen from a set of two gold codes. The gold codes have low cross correlation and autocorrelation features. The gold codes are generated using two feedback shift registers and have a specified sequence of pseudorandom integers. G1 and G2 are two 10-bit shift registers that generate the longest pseudo noise (PN) codes, with a length of 1,023 bits. The block diagram of code generator circuit of C/A code is shown in Table (2.2) [2]

Table 2. 2 GPS code generator polynomial and initial state

Register	Polynomial	Initial state
C/A Code G1	$1 + X^3 + X^{10}$	1111111111
C/A Code G2	$1 + X^2 + X^3 + X^6 + X^8 + X^9 + X^{10}$	1111111111

As noticeable from the Table (2.2) [2] The exclusive OR of the delayed release of the G2 output sequence and the direct output of the G1 sequence is the unique code for every GPS satellite. The C/A code is free of charge and only transmitted on the L1 frequency for all GPS users. For the GPS C/A code generator design, there are 1023 potential gold codes, while using exclusive ORs of two maximum shift registers, as illustrated above. Therefore, only 37 gold codes with best properties have been picked for the GPS space segment, so all sequences have not low correlation properties. The GPS receiver must differentiate between transmissions from 12 satellites using correlation techniques, therefore low cross-correlation of sequence is essential [2]. This important property of the

GPS signal plays important part in the simulation to detect the targets with the help of reflected GPS signals.

2.1.7 The P-Code

The P-code is a 10.23 MHz PRN code sequence that have a length of 267 days. The P-code is made for military purposes so, it is encrypted and available only to authorized users. The P-code is beyond the scope of this research so it will not be discussed further.

2.1.8 Navigation Data Structure

The navigation data message of GPS is modulated onto L1 (1.58 GHz) and L2 (1.23 GHz) carrier signal using Binary Phase Shift Keying (BPSK). The navigation message consists of five frames and each frame consists of 1500 bits with a transmission rate of 50 bits per second. The transmission of one frame lasts about 30 seconds (1 navigation bit per 20 C/A code sequences). Each frame is divided into five equal length sub-frames that are numbered from 1 to 5. The satellite transmits information about its position and velocity, transmitter clock biases and drifts and ionospheric correction parameters in sub-frame 1 to 3. The almanac information about the full GPS constellation is in sub-frame 4 and 5. The sub-frames 4 and 5 are sub commutated 25 times in order to accommodate almanac data for full GPS constellation. Therefore, there are 25 different versions of sub-frames 4 and 5 and it takes about $25 \times 5 \times 6s = 12.5 \text{ min}$ to complete a full navigation data cycle [15].

Each sub-frame consists of 300 bits that are further divided into 10 words of 30 bits each. The first 24 bits in each word are data bits followed by 6 parity bits. The purpose of these parity bits is to protect the data message against bit errors. The first two words in each

sub-frame consists of a characteristic marker bit pattern Telemetry Word (TLM) and a time tag Handover Word (HOW). There exists a high probability that the full navigation data cycle of 12.5 min repeat unchanged except for the updated time tag information encoded into HOW [15].

2.1.9 GPS signals Modernization

A major focus of the GPS modernization program is to add new navigation signals to the satellite constellation. Three new signals have been designed for civilian users: L2C, L5 and L1C. The legacy civil signal L1 C/A or C/A at L1 will continue its broadcast, for a total of four GPS signals. These new signals are summarized as [16]:

L2C (1227 MHz): L2C is designed specifically to meet commercial needs. L2C enables ionospheric corrections when combined with L1 C/A in a dual frequency receiver, a technique that boosts accuracy. L2C enables faster signal acquisition, greater operating range and enhanced reliability for professional users with existing dual frequency operations. L2C makes it easier to receive under trees and even indoors because, it broadcasts at higher effective power than the legacy L1 C/A signal.

L5 (1176 MHz): L5 is designed to meet demanding requirements for safety of life transportation and other high performance applications. This signal broadcasts in a radio band reserved exclusively for aviation services and features high power, wider bandwidth and an advanced signal design. The use of L5 signal in combination with L1 C/A for aircraft will improve accuracy (via ionospheric corrections) and robustness (via signal redundancy).

L1C (1575 MHz): L1C is designed to enable interoperability between GPS and international satellite navigation systems. This signal features a Multiplexed Binary

Offset Carrier (MBOC) modulation technique that enables international cooperation while protecting U.S national security interests. The L1C is designed to improve the mobile GPS reception in cities and other challenging environments [16].

2.1.10 C/A Code Correlation Features

The integration of the received signal with the waveform duplicate is termed correlation. Basically, it is a statistical process to measure the similarity. The purpose of computing correlation is to measure which two signals are similar. This process is implemented in receiver and used to find PRN code changing for each satellite. There are two types of correlation: one is auto-correlation which is used to measure the relationship within the signal itself and other is cross-correlation which is used to measure the similarity between two different signals [17].

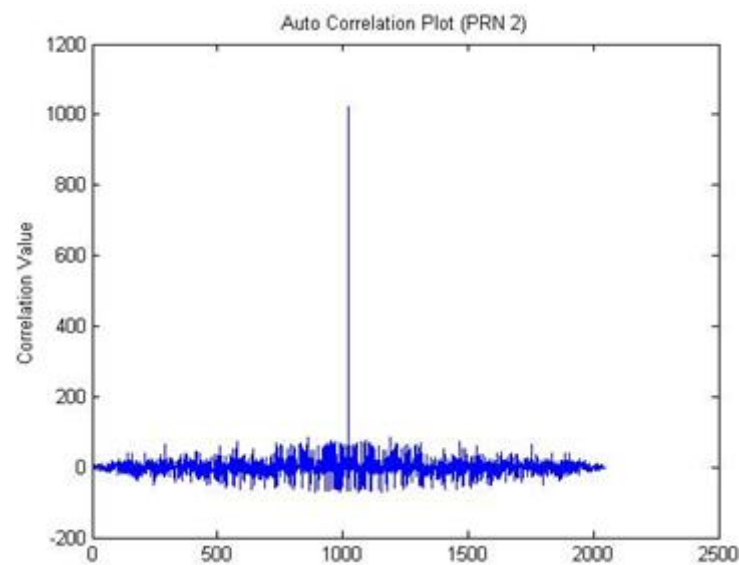


Figure 2. 2 Autocorrelation of C/A code

The correlation peak value is far more significant than the other correlation values, as seen in figure (2.2). In the auto-correlation figure, the secondary peaks are at least 24dB lower than the higher peaks. The maximum correlation value is 1023, which is the

number of chips in one C/A code set. This critical characteristic of the GPS signal is important in the simulation used to detect targets using reflected GPS signals.

2.1.11 Pseudorange Determination

The GPS satellites deliver signals to the receivers for measuring the arrival time (TOA). As the light rate is known, the distance from the delay may be estimated.

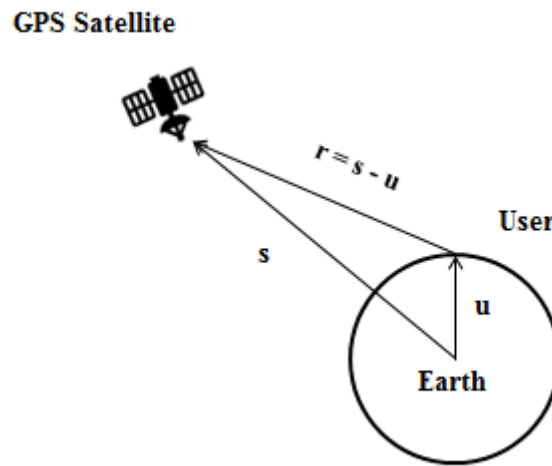


Figure 2. 3 A Depiction for Calculating ‘r’

The position of the satellite s is computed in ECEF coordinates using satellite ephemeris data. If a user is positioned at ' u ', the user to satellite vector is ' r ,' as shown in Figure (2.3) [18]

$$\mathbf{r} = \mathbf{s} - \mathbf{u} \quad (2.1)$$

The distance between a satellite and the user is denoted by the ' r .'

$$r = \|\mathbf{r}\| = \|\mathbf{s} - \mathbf{u}\| \quad (2.2)$$

The distance ' r ' is calculated by calculating the signal propagation time from the satellite to the receiver. If the signal's broadcast time is T_s and the signal's receiving time at the receiver is T_r , the geometric distance ' r ' may be represented as:

$$r = c (T_r - T_s) \quad (2.3)$$

Where 'c' is refer to as speed of the light.

2.1.12 GPS Errors

In GPS navigation, error sources may be split in two groups, bias and noise. The noise errors include receiver noise while bias errors includes errors in the pseudorange measurements that degrades accuracy of the system [19]. These bias errors are described below:

Table 2. 3 Definitions of GPS errors

Type of error	Description
Ephemeris data	Errors in the transmitted location of the satellite
Satellite clock	Errors in the transmitted clock
Ionosphere	Errors in the corrections of pseudo range caused by ionospheric effects
Troposphere	Errors in the corrections of pseudo range caused by tropospheric effects
Multipath	Errors caused by reflected signals entering the receiver antenna
Receiver	Errors in the receiver's measurement of range caused by thermal noise, software accuracy and interchannel bias

These errors are briefly described below:

2.1.12.1 Ephemeris Error

At 22,000 km, the GPS satellite is susceptible to cosmic influences such as solar winds and the Earth's gravitational pull which have a negative effect over satellite trajectory. These forces could not be estimated due to which they caused an error in the calculation of satellite position derived from the ephemeris data. As compared to other errors,

ephemeris errors do not have a significant impact on pseudorange measurements and receiver tracking performance. The ephemeris errors may have some effect on location of the target as ephemeris data is used to calculate the satellite position during image reconstruction.

2.1.12.2 Satellite clock error

The GPS satellite uses atomic clocks (cesium or rubidium) that are very stable. The stability of the clocks is about 1 part in 10^{13} over a day. However, the control stations on ground are continuously monitoring the satellite clock deviations and upload the correction parameters. These updated parameters are then transmitted to GPS receivers through satellite broadcast data and receivers can use this information to compensate the satellite clock errors.

2.1.12.3 Ionospheric error

Ionosphere consists of invisible layers of ions and electrons that are suspended in the Earth's atmosphere about an altitude of 50 Km's and above. The Sun's ultraviolet light is main source of these layers that ionizes atoms and molecules in the Earth's upper atmosphere. The ionospheric delay is the main problem in achieving precise positioning and this error can bias the receiver user's position up to 15m from the actual coordinates [20]. The civilian users can only estimate the delay by using an ionospheric model. The Klobuchar model is used for single frequency GPS receivers that accounts for about half of the day.

2.1.12.4 Tropospheric error

For frequencies as high as 15 GHz, troposphere, which is low in the atmosphere, is neutral electrically and non-dispersive. The group and phase velocities of GPS signals on both L1 and L2 frequencies are equal within this medium. The resulting delay is a

function of atmospheric temperature, pressure and moisture content, which induces pseudorange and carrier phase errors from about two meters for a satellite at zenith to more than 20 meters for a low elevation satellite. In real time, it is impossible to predict the error introduced by troposphere, being affected by the weather conditions between the receiver and satellite (the amount of water in the line of sight to the satellite being the major factor) [21]

2.1.12.5 Multipath error

Multipath error of GPS signals are subject to the reflection and diffraction like any other source of electromagnetic waves. This is caused by the reception of signals arrived not only directly by the satellite but also reflected from the objects on the surface of earth. This error results from interference between two radio waves that have traveled paths of different lengths between the transmitter and the receiver [22]. They will not be discussed further in details as we are exploiting this property of GPS signals to locate the target in the region of interest.

2.1.12.6 Receiver error

Receiver noise errors can be divided into two different categories. The first one includes the AGC finite response time shifts, sampling, quantization losses, loss in the code tracking design and hardware design limitations while the second one originates from the practical limitations of component accuracy. These errors include receiver oscillator error, filter induced distortions on the signal and inherent system noise. These errors are not completely minimized due to complexity and cost limitations and they are more related to the hardware implementation.

2.1.12.7 Error in Receiver's clock

The offset between the receiver clock and the GPS system time is known as receiver clock error. The receiver clock error leads to a range error in the pseudorange and carrier phase measurement. Similar to the satellite clock error, the receiver clock error can be eliminated by differencing between the observations observed in a receiver between two different satellites [23].

2.1.12.8 Dilution of Precision (DOP)

The major changes are caused by geometry called dilution of precision (DOP). It has been revealed that satellites with an overhead and a lower horizon scattered equally around the receiver are preferable. This effect can be illustrated by the figure (2.4).when satellites are clustered together while geometric dilutions of 2.5 are about the [24]

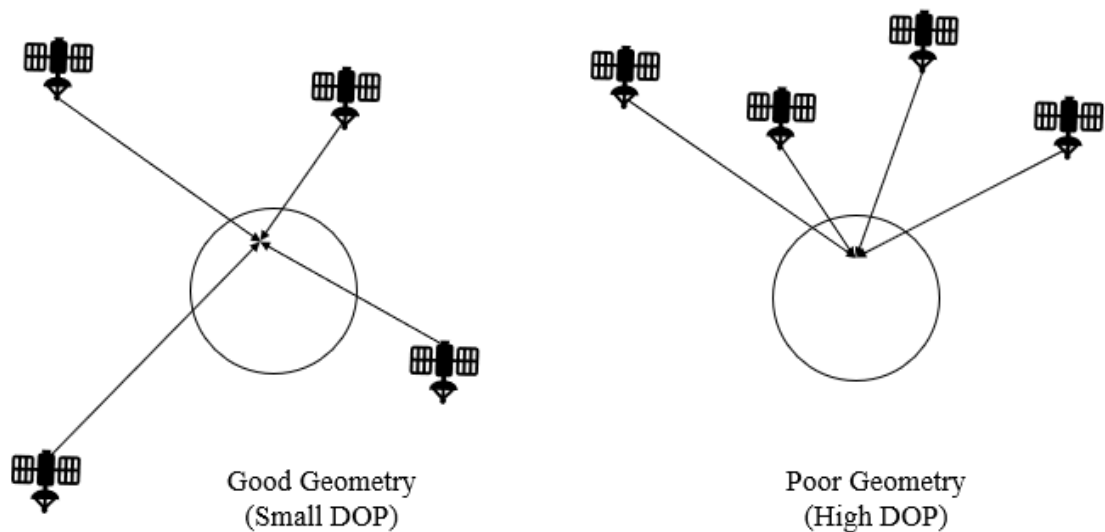


Figure 2. 4 Comparison of small DOP and high DOP

2.2 Doppler Frequency

Whenever a receiver is in motion relative to the GPS satellite, the received signal will have frequency different from that of transmitted signal by a Doppler frequency offset. The ephemeris information and orbital model that is present in the receiver helps to compute the velocity of satellite. The frequency of receiver changes as the satellite move towards the receiver or when it is moving away. When the satellite is at the closest position relative to the receiver then the Doppler frequency shift of zero occurs. The Doppler frequency shift caused by the satellite motion affects both the carrier frequency and the C/A code. The maximum change in the Doppler velocity (v_d) over time can be computed to approximately 0.178 m/s^2 and the maximum rate of change of the Doppler frequency comes out to be:

$$\delta f_{dr}|_{max} = \frac{dv_d}{dt} \frac{f_r}{c} = \frac{0.178 \times 1575.42 \times 10^6}{2.99792458 \times 10^8} = 0.936 \text{ Hz} / \text{s} \quad (2.4)$$

This value is very small which allows the tracking loop to be updated in every second. It can be seen that the rate of change of the Doppler frequency caused by the satellite's motion is very low and the update rate of the tracking program is much more frequent for the satellite motion to have any significant effects [25].

2.3 Matched Filter

A matching filter helps to maximize the instantaneous relationship between peak signal power and mean noise power. When changing the signal, the filter needs to be updated to match the signal used. The aim of the matching filter is to offer the greatest signal-to-noise ratio for transmitted signal at its output [26].

2.4 FM Chirp Signal

The waveforms are selected on their ability to satisfy the requirements for detection, measurement accuracy, resolution, ambiguity and clutter rejection in the radar systems [27]. In a radar environment, long duration pulses are required in order to increase signal energy for improved detection [28]. While short duration pulses are required to improve target resolution (a term that is used to imply that two or more targets are separated to perform necessary measurements and is defined as the minimum distance on the ground at which two object points can be imaged separately). A waveform can be designed with the help of Frequency Modulation (FM) to have both long duration and small effective duration. The linear frequency modulation (LFM) pulse is called chirp signal and this can be represented as:

$$f(t) = \cos(\omega_0 t + \pi F/T t^2) \quad \text{for} \quad -T/2 < t < T/2 \quad (2.6)$$

Where $F = f_2 - f_1$ is the sweep bandwidth with f_1 and f_2 being the initial and final frequencies, where 'T' is the time duration as shown in figure (2.6).

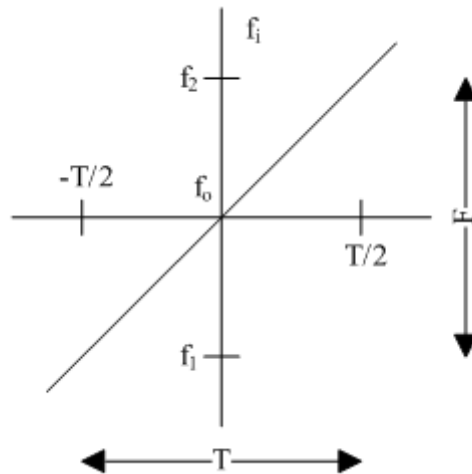


Figure 2. 5 Chirp signal

2.5 Target Characteristics

The power density ratio of the signal to the receiver is known as the Radar Cross Section (RCS) of the object referring to the power density of the radio wave event. Consequently, RCS measures the ability of the target to display radar signals in receiver direction. The RCS unit is square meter and the signal-reflected measurements are made in comparison to the signal reflected by a completely smooth body with a sectional area of 1 m².

The RCS can also be expressed as:

$$\sigma = 4\pi r^2 \frac{P_s}{P_i} \quad (2.7)$$

Where P_i is incident power density at a scattering target from a distance radar and P_s is the scattered power density [29]. When the signal reaches the target, it is reflected in ways which rely on the toughness of the surface and the di-electricity of the target. These signals that do not reflect the various effects of imaging are most valuable. In such kind of 'reflection' signal is dispersed across a broad angle and is therefore further attenuated in accordance with the distance.

A multitude of 'point reflectors' can be regarded as the surface of the complex target. In reality, the 'point reflector' must be of a finite size and it seems appropriate to model the 'point reflectors' as diameter areas comparable to the imaging approach for our purposes. According to the inverse square law, the signal reflected in a 10 cm radius would attenuate at a rate of around $20 \log_{10}(R/0.1)$ (expressed in dB) where 'R' is range from the point reflector to the point of observation [30].

A target can be resolved from another target if it falls into a different bin or cell and the return is not marked by self-clutter [31]. A separate response will be generated when two scatters of comparable strength falls within different resolution cells. It is assumed in the

simulation that the targets act like point targets and do not experience power deterioration due to their RCS nor do they depend on the angle of incident of the reflected signal. There is an advantage of using point like scatter characteristics that gives much less geometric de-correlation.

2.6 Radar Resolution

The resolution of radar is the ability to detect and distinguish closely situated targets and to measure their coordinated and velocity separately [32]. The factors effecting radar resolution of targets in range, velocity and angular position are as follows:

$$\text{Range Resolution } (\Delta R) = \frac{c \tau_{out}}{2} \quad (2.8)$$

Where ‘ c ’ is the velocity of light and τ_{out} is the duration of echo pulse. The target range is indicated by the Doppler frequency shift. The radar resolution in target velocity is expressed as:

$$\Delta V_R = \frac{\lambda}{2\tau} \quad (2.9)$$

Where ‘ λ ’ is the wavelength of radar signal and ‘ τ ’ is the time interval. The angular resolution of radar depends upon the antenna system beam width in the relevant plane.

2.7 Radar’s Ambiguity Function

The ambiguity function and its plot provide a qualitative assessment of different waveforms in meeting radar system requirements in terms of target resolution and clutter-rejection capabilities. The response of the matched filter radar receiver to a target

displaced in range delay τ and Doppler frequency ν from a reference target is known as radar ambiguity function. The response is measured at a time instant when the reference target is at its maximum value. The ambiguity function is closely related to the matched filter output waveform. The matched filter output due to a Doppler shifted echo is given by:

$$g(t) = \int u(x)u^*(x-t) \exp(j2\pi\nu x) dx \quad (2.10)$$

Where (*) represent the complex conjugate. The time reversed matched filter response which is also referred to as the correlation function of $u(x)$, is obtained from equation (2.10) by substituting $-\tau$ for t .

$$\chi_u(\tau, \nu) = g(-\tau, \nu) = \int u(x)u^*(x+\tau) \exp(j2\pi\nu x) dx \quad (2.11)$$

The ambiguity function of radar is defined as the square magnitude of the time reversed matched filter response.

$$\psi_u(\tau, \nu) = |\chi_u(\tau, \nu)|^2 = \left| \int u(x)u^*(x+\tau) \exp(-j2\pi\nu x) dx \right|^2 \quad (2.12)$$

Where ' τ ' is the range delay, ' ν ' is the Doppler frequency and $u(x)$ is the transit waveform. The resolution of SAR is analyzed with the help of ambiguity function, based on equation (2.11).

2.8 Synthetic Aperture Radar (SAR)

Synthetic aperture radar (SAR) is based on signal processing to provide a long effective antenna rather than a lengthy antenna.. In most instances, only one radiation element is used. The antenna is translated in successive positions along one line and a signal in each

line is transmitted. As illustrated in figure (2.7) radars signals received as a result of this transmission are stored [33]. The storage needs to be done to maintain the amplitude and phase of the incoming signals.[34].

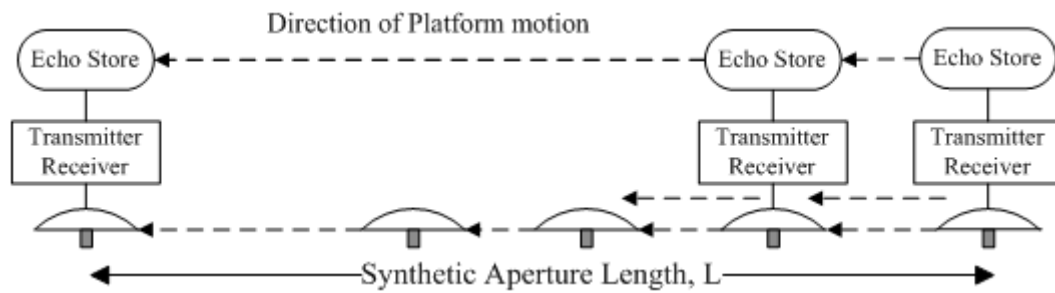


Figure 2. 6 Construction of a synthetic aperture

After the radiating element has travelled a distance, the signals in storage closely mirror the signals received by the elements of an actual linear array. If the signals in storage are exposed to the same techniques as those used to build a physical linear array, the effect of a large antenna aperture can be obtained. As a result of this concept, the term "synthetic aperture" has been coined to describe this procedure [34].

There are two types of SAR's, one is mono-static radar in which we use same antenna for transmission and reception while other type is bi-static radar in which we use separate antennas for transmission and reception purposes. The principle of bistatic radar is based on a radar mounted on an aerial platform. The transmitter and receiver may be on the same platform, whereas the target item may be stationary or moving [34].

2.9 Bistatic Radar

Bistatic radar is a form of radar that transmits and receives signal using antennas in various locations. As depicted in figure (2.8) [35] a transmitting and receiving antenna is placed at different sites, separated by a distance 'L' which is called baseline or baseline range. We can locate the target on another site. Anyone of the mentioned sites could be in space or on earth and can be stationary or moving [36].

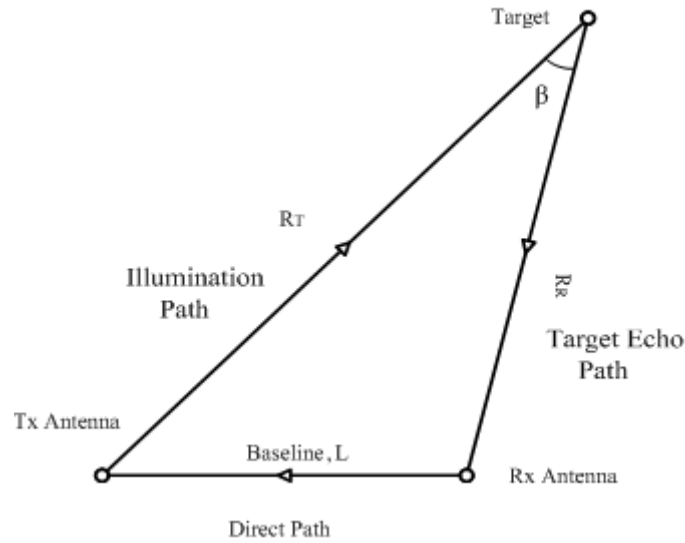


Figure 2. 7 Bi-static geometry

The angle formed by the transmitter and receiver with the vertex at the target is known as the bistatic angle, and it is denoted by (β). This parameter is used in many bistatic equations and is unique one. It is essential that the transmitted time of signal must be known to the receiver. This measured time is converted to range sum estimate ($R_T + R_R$), where R_T represents “transmitter to target” range and R_R represents the “target to receiver” range.

In case of a mono-static radar $R_T = R_R$, while in case of a bistatic radar $R_T \neq R_R$ in nearly all cases. Therefore, the bistatic radar must solve the transmitter-target-receiver triangle to obtain an estimate for R_T or R_R . An estimate of the transmitter's position with respect to the receiver is required in this solution. The basic requirements for the bistatic radar can be solved by considering GPS as the transmitter of opportunity. But one of the major issues is still the appalling signal to noise ratio of reflected GPS signals.

3 LITERATURE REVIEW

3.1 Initial Research regarding GPS Imaging

An initial research with regard to an imaging system based on a 3D multistatic SAR system that utilizes reflected GPS signals from different objects that were located on the surface of Earth was documented by Chris Rizos and co-authors in [37]. For detecting the movement and changes of the imaged object, the principle of bistatic radar was used. A matched filter was used to process indirect GPS signals in order to enhance the detecting spatial resolution. The imagesystem was thereafter determined by measurements of spatial resolution and confirmed by simulations using Matlab. Researchers at a given receiver and object position were considering several scenarios for the visible satellite. The simulations confirmed that the spatial detection resolution is dependent upon the relative position of the GPS satellites, the optical objects and the velocity. The authors employed a matched filter method to improve the system resolution during signal processing. The filter output was as follows:

$$S_{ij}(\tau) = \int_{-\frac{T_s}{2}}^{\frac{T_s}{2}-\tau} S_{Rij}(t+\tau) S_{Rij}^*(t) dt \quad (3.3)$$

Where T_s represents the time of observation and $(*)$ is the complex conjugate.

The resolution is directly related to the relative velocities of the object and receiver as shown by simulations. The higher resolution is the result of greater relative speed. This was demonstrated by the thesis author during simulations that Increased speed causes greater change in the geometry, resulting in improved resolution. It was also demonstrated that spatial resolution (ρ) is a function of the transmitter's position with regard to the object and receiver. The results and finding presented in [37] conform to the results presented in this thesis and many findings are similar. The scientists have not carried out further study with the aim of using new algorithms for reconstruction.

The thesis author is pleased to demonstrate that from realistic observation distances, theoretically, a spatial resolution (19 cm) for the L1 wavelength (about 2-3 hours in static receiver) is achievable provided that adequate coherent integration time is permitted and direct signals are suitably suppressed. The reconstruction procedure led to an unacceptable PSF that substantially blurred the recovered image.

B. Mojarrabi et al. examined a power budget analysis for passive target identification and imaging employing reflected GPS signals, which is another application of GPS signals. [38]. Their study concentrated on the low power levels of the GPS signal near the Earth's surface (about -160 dBW). Because of the structure of the reflector, they predicted that some power would be absorbed, resulting in a further loss in signal intensity. An important account in the study concerned the highest detection range R_{max} , given by:

$$R_{max} = \frac{1}{\rho_o} \left(\frac{A_e}{\lambda} \right)^2 \frac{G}{\Delta F} \frac{1}{\tau_k} \quad (3.4)$$

Where ρ_o is the signal to noise ratio, A_e being effective antenna aperture, radar cross-section is σ and receiver bandwidth is ΔF . The target coherence time interval is given by τ_k and GPS wavelength and antenna gain is given by λ , G respectively. It is possible to evaluate the maximum target detection range, by putting the necessary values in above-mentioned equation, the maximum detection range R_{max} can be calculated to be about 214 km. There was no further research activity available by the same authors in this area. However it was revealed that some groups are busy conducting research in the relevant field and their work will be explained further in this chapter.

3.2 Types of GPS Remote Sensing

The reflected GPS signals provide opportunity for different applications that are used for remote sensing other than imaging. In 1993 the European Space Agency (ESA) presented the concept of the use of GPS signals from the earth's surface as a remote sensing system [39].

This concept has also been examined in a number of research papers. These research papers discuss different aspects of GPS remote sensing, most of them are relevant to the collection of GPS data for different kind of remote sensing applications. The research that is most relevant has been collected and reviewed. The data can be used for the following applications:

- a) To calculate mean sea height over ocean, significant wave height, wind speed and direction estimation.
- b) Over ice to measure ice thickness, surface ice density and ice age.
- c) Over land for measuring the content of soil moisture, biomass and imagery.

d) Using SAR images for early warning system for oil spill.

At least one example of every application listed above has been attempted to explain by the author.

The research for remote sensing applications exploiting GPS reflected signals begun in 1996 by Katzberg and L Garrison at the NASA Langley Research Center (LaRC). The study leads to a series of publications describing this technique's theory and mechanism. The authors estimated wind speed measurements using forward scattered GPS signals [40]. This research was based on "Effect of sea roughness on bistatically scattered range coded signals from the global positioning system" This work presents a technique which measures the cross-correlation form between a reflected signal and a local PRN code. For this experiment, it was employed to follow the direct path of satellites via a right handing circularly polarized antenna (RHCP). To capture the cross correlation function from reflected GPS signals, a system using a nadir-oriented left-hand circular polarized antenna (LHCP) has been employed. A method of inversion of bistatic scattering models was applied for the estimation of wind speed from correlation measurements, which enabled the calculations for sea surface roughness to be performed, which in turn is known to be dependent upon the surface wind speed. The distribution form of signal power was compared to analytical models using a geometric optical methodology. Two strategies have been studied to match these functions. The initial technique recognized that the most important information contents in the reflected signal are located near the waveform's trailing edge. By getting the non-linear lowest-square estimation, the second sought to conform to the completed shape of the waveform. This study successfully relies on the form of the wavy correlation function collected from the forward scattering GPS signals to rely on geophysical data.

Another important application of remote sensing is the study of surface reflected signals to evaluate polar sea ice conditions. [41]. The environmental characteristics of this region make this application more useful. As we already know, GPS signals are available 24 hours a day and can be captured for later signal processing. In this research a system containing a GPS receiver, two GPS antennas, processor, some cables and a data recorder device were used. The GPS receiver was modified by means of a nadir LHCP antenna to measure the correlation power of the offset delay and Doppler values. A reflected signal at various delays and Doppler values was the first measurement.

Cinzia Zyffada and Valery Zavorotny further investigated the coherence of the time; statistical features of GPS reflected signals and effects on accuracy of remote sensing of the sea and surface topography. The researchers have stated that when a GPS signal is influenced on the surface of ocean then it is scattered back in all the directions. This dispersion contributes the most towards the forward dispersion of the specular point of reflection. An ellipse that stretches from the transmitter to the receiver at any time during monitoring is formed by a specified area of the circumference. The effect of the atmospheric aspects, the geometry of the system, and the parameters can be in the waveform of the received power to be investigated after combining simultaneous surface contributions to the receiving antenna. Initial studies in observing GPS reflections in Arctic sea ice revealed that for moderate altitudes of the GPS receiver in air, the shape of the reflected signal was pretty constant (sharp, narrow waveform throughout the flight), which indicates that differences in ice surface roughness were not significant in the L band. The peak power, on the other hand, varied greatly along the flight path. The maximum strength of the received GPS signal changes with the reflective surface, and the researchers concluded that, in addition to the basic ability to identify the presence of sea ice, a GPS signal could provide useful information regarding ice conditions [42].

Some other experiments were also taken into account in which this equipment was installed on low Earth observation satellites called LEOs. The general depiction of using reflected signals from a space platform is shown in figure (3.1) [43].

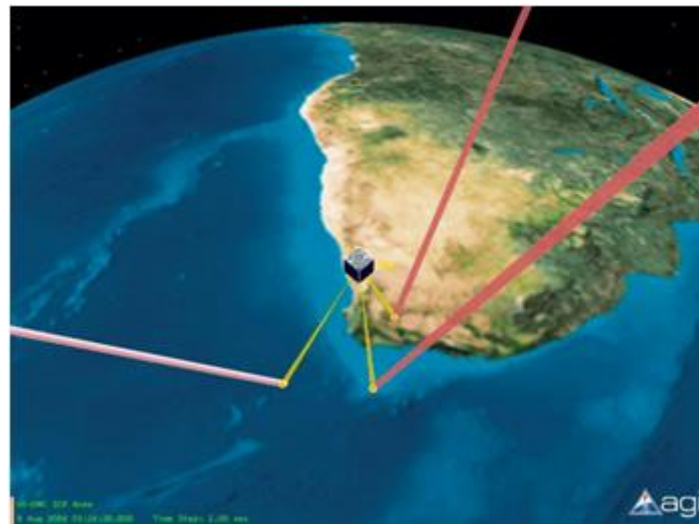


Figure 3. 1 Bi-static reflection geometry for signals from three GNSS satellites received in LEO

In October 2003, The British Disaster Monitoring Satellite Constellation (UK-DMC) was launched which made it possible to test the feasibility of monitoring reflected signals from LEO orbit. This is one of the small and low cost satellites built by Surrey Satellite Technology Limited (SSTL). The main purpose was to provide daily images for different kind of applications that also includes global disaster monitoring. The experimental results have shown that this is a workable technique for remotely sensing the ocean roughness and it has great potential to make useful scientific measurements in many other research areas, including Earth and ice sensing applications [43].

Another effort was carried out by Student Reflected GPS Experiment (SuRGE) that was basically a university class scientific mission to measure the GPS reflected signals [44]. The

reflected signals have ability to exhibit information of physical properties about the reflecting surface. The signal delay measurement is a characteristic of the sea level height, the delay distribution and the intensity of the Doppler signal that provides information about the roughness of the sea, mainly caused by waves driven by the wind.

This research also examined the possibility of recovering soil moisture from GPS reflections of the Earth's surface. This study's instruments include a receiver for real-time tracking and delay Doppler mapping, as well as an S-band analogue signal interpreter for transmitting a complete analogue GPS signal for recording at particular ground stations. The reflection data from the GPS was down converted and retransmitted for post-processing and restoration of land and ocean surface properties. This project presented an excellent opportunity for NASA to investigate a new approach for remote sensing of oceans and land surfaces utilizing GPS, as well as extraordinary opportunities for student participation among several institutions and aerospace firms, including NASA.

An experiment has been proposed to illustrate the data collection of wind speed, direction and altitude measurements at the ocean surface from received GPS signals that are bistatically scattered in the LEO orbit [45]. The Monte Carlo simulation method was used for predicting the statistical errors in satellite based wind speed monitoring by using bi-static GPS signals. For simulating the mean waveform after correlation, a scattering model based on geometric optics and experimental anisotropy of the ocean surface was used and analytical simulations were conducted to predict the performance of these measurements. The errors due to statistical noise were simulated in the system and the results proved that they could be reduced to 1 m/s or less.

In another research by S. Katzberg et al. a new technique was described for measuring soil moisture using bi-static GPS radar [46]. For the remote sensing of soil moisture, an ideal frequency is 1.57542 GHz of GPS L1. The results have demonstrated the temporal change in the surface reflectivity by using GPS bi-static radar measurements. The strength of the dispersed signal measured using the data sampled in situ soil moisture at 31 different field sites was analyzed. It was founded that bi-static GPS radar data exhibited positive in earth moisture trends, from dry to moist. The dipole dispersion was theoretically modeled and confirmed that the amplitude and width of the waveform reflection depend on its dielectric permittivity, covered vegetation and soil roughness.

Because, the GPS signal is available all day long under all-weather circumstances, this method is more convenient for wide spread applications with low cost GPS receivers compared to other remote sensing systems [47]. Multiple delay signals with amplitudes proportional to the power reflection factor will be produced at each reflection point by reflection of the GPS signal from the rough insulating surface. A typical GPS reflection receiver gives an output of cross correlation between a PRN code of reflected signal and a replica of PRN code generated by a receiver. This cross correlation function conveys information about the roughness and dielectric characteristics of soil at specular locations and is typically larger and lower than the auto correlation of the direct signal. The GPS receiver was modified and was kept on a stationary tower of altitude 300m. The stationary tower's monitoring design enables the measurement of GPS reflections from the same terrain area, hence excluding signal abnormalities caused by variations in the roughness and vegetation sceneries. This experiment was conducted for many days and four local rain fall events were also happened during this experiment. The reflected signal was analyzed and an expected increase in the intensity of the reflected signal was shown to diminish following rain events

and to increase during consecutive droughts, thus demonstrating the dependence of reflected signals on soil moisture content [47].

3.3 Atmospheric Effects on GPS Signals

As discussed earlier in this thesis, the atmosphere, including the ionosphere and troposphere, has a great influence on the propagation of GPS signals. The ionosphere causes an error in pseudorange by the signal delay and effect position calculations. These errors can be reduced by using an ionospheric model which estimates the delay. In the case of a single frequency receiver, which accounts for roughly half of the delay, the Klobuchar model is used. This model takes into account factors such as sunspot activity and total electron count (TEC). The almanac is the fourth sub frame of the navigation message in which mandatory data is included. The ionosphere is treated as one layer, situated around 350 kms above the surface of the Earth using a thin shell approximation in Klobuchar model. Klobuchar Model needs some knowing of geodetic latitude and longitude, elevation angle, and azimuth for each GPS satellite in order to calculate the ionospheric delay. The test results have stated that 50% to 60% of the delay due to ionosphere can be compensated by this model in the middle latitude geographic range, thus satisfying the general non-precision navigation applications [48].

Another interesting fact about GPS signals reflected from the ocean surface is that, under certain conditions, it may be used to calculate the TEC of the ionosphere at a satellite placement. Determining the TEC and applying the data to ionosphere models can improve their accuracy in places where it is most needed. The use of a GPS reflected signal in the ocean by adding a GPS receiver and downstream aerial antenna to satellites with single-frequency altimeter is briefly discussed by Stephen J. Katzberg and his co-authors [49].

3.4 Bistatic Radar

The bistatic radar has already been discussed in this thesis. However, it would be judicious to provide some historical account of bistatic radar at this stage. Since the early days of radar, bi-static radar systems have been investigated and developed. [50]. For example, during World War II, Germans used British Chain Home Radars as lighting systems for their bi-static systems in Klein Heidelberg. Currently, there is a notable interest in passive coherent location (PCL) technologies, which use transmission and communication signals as “hitchhiking”. There are many reasons to be positive about the subject and a strong possibility that bistatic radar systems may be developed and employed in the near future [50].

For instance, there is more spectral interference. Military operations are anticipated to take place near densely populated areas, where there are many transmission and communication signals. This spectral obstruction is a problem for most of the purposes but fortunately for PCL it can be exploited in its favor. Furthermore, the very-high frequency (VHF) and ultra-high frequency (UHF) frequencies used in high-power FM radio and television broadcasts are suitable for PCL in many aspects. As it is already mentioned, bi-static receivers are more likely to be simple and inexpensive. With the advent of GPS, several timing issues that previously restricted the performance of bi-static systems have been resolved. Last but not least, significant advances in signal processing capacity mean that many signal digitization and processing processes may now be performed in real time [50].

It has been further explained that the properties of 'transmitters of opportunity' may be assessed on following terms: spatial and temporal coverage, power density at the target and waveform. By digitizing off-air waveforms and calculating and mapping their functions and ambiguities, the waveform characteristics of a number of PCL luminaires (VHF FM radio, analogue and digital television (TV), and digital audio transmission (DAB)) have been studied in [51]. It is also possible to employ a Global System for Mobile Communications

(GSM) base station transmitter as an opportunity illuminator. The work done in [52] demonstrates the potential capabilities of this type of passive radar for tracking of sea and air moving targets and for the Doppler frequency detection.

Detection of air targets by using GPS as a bistatic radar illuminator has been documented in [53]. In this research, the difference in radar cross section of an air target and reflections from ground disturbances is discussed that is followed by an estimate of the effect on the performance of such a system of ground disturbances. This research reproduces the analysis of [54] by using a conventional approach but with different selection of parameters, which results in less optimistic estimates of the maximum detection range. This research concludes that due to the very weak transmitted GPS signal, it is probably not suitable to detect targets, with a small RCS using receiving antennas with 15 dB gain. Some other problems to be overcome are related to the "near-far" problem resulting from the dynamic range of the received GPS signals and the problem that the received background noise power has a power level greater than the reflection from the target. During execution of the experiment for image generation, the thesis author has also noticed excessive ground clutter, suspected to be generated by the reflected signal from the university building and other objects behind the target.

For these difficulties, the construction of such system can be implemented by following some suggested recommendations. Firstly, the gain at the receiving antenna should be as large as possible. Secondly, extremely long coherent integration periods using full data-wiping should be employed because of the appalling signal to noise ratio of reflected signal [53].

A method for non-coherent combination of the outputs of all available satellites can also be implemented. As we know that, usually six satellites are available and sometimes this number reaches to as many as twelve that could make it possible to increase the maximum range by a

further factor of 1.5 to 3, depending on the squaring loss. This point can be taken into consideration during the next phase of research in order to improve the image quality, however, it may result in undue interference due to multiple satellites.

3.4.1 Resolution of Bistatic Radar

A remote sensing system is built around the key parameter of spatial resolution. Many research papers and articles discuss the resolution aspect of BSAR. In a SAR system, the spatial resolution is characterized by the Point Spread Function (PSF) [55]. In other research the performance of bistatic SAR system has been evaluated to select areas on the ground that meets the requirements for range and cross range resolution [56]. To derive the equations of range and cross range resolution, Doppler resolution in meter, vector gradients have been used. The area where the scope and the cross-sectional resolution are less than a meter was shown in computer simulations of a transmitter at an altitude of 1000 km, with an integration time of 10 seconds and a 500 MHz radar bandwidth.

The authors have argued in [57] about the equations that they can only be used on range signals with a rectangular spectral shape. They briefly introduce spatial BSAR generalized ambiguity functions (GAFs), which they depict initially in the Doppler domain before extending to the spatial domain (coordinates). With this GAF spatial representation, The azimuth and range resolutions, as well as the resolution cell area, are thought to be related to various directions and coordinate systems.

The thesis author's computations demonstrate that a spatial resolution of the order of wavelength L_1 (19 cm) is attainable if a sufficient coherent correlator integration time (at least 2 hours in the case of a fixed receiver) is permitted and the direct signals are suppressed appropriately. That is a novel accomplishment and considered a very significant progress in

the field of remote sensing to achieve a sub meter level resolution with an imaging system that utilizes GPS reflected signals without the availability of dedicated transmitter. This amazing achievement has been made possible with the help of efficient reconstruction algorithms developed by the thesis author, which will be further elaborated in the next chapter. However, the resulting image had severe blurriness, which was caused by the low quality of the PSF, which was improved using a deconvolution technique based on the Wiener filter.

There are many possible applications for this type of imaging device, especially for monitoring of land surface. However, this is understood that the practical implementation of this system would be very challenging and to achieve similar resolution in a real world environment, a very concerted effort will be required. Some of the aspects and factors that can have an adverse effect on the resolution and image quality are the weak reflected GPS signals' poor signal to noise ratio, radar clutter and back scattering, hardware limitations, GPS signals available for longer periods of time and other pertinent problems. After successful simulations, the aim of further research endeavors and efforts is to bring the resolution and image quality of an actual image as close to the simulated image as possible.

3.5 Other Relevant Research

It would be unfair to not mention the work done by research group of Dr. M Cherniakov from Microwave Integrated Systems Laboratory at the University of Birmingham. In general, this group has done outstanding work in the field of bistatic radars (BR) and, in particular, the use of diverse systems as a broadcaster of opportunity. In a first report [58], this research group studied a system that was aimed at analyzing the ability to detect airborne targets using bistatic static radars and identify the problems associated to it. The parameters used in this

research were from Iridium LEO satellite system and the focus was to detect airborne targets in a context of white noise, clutter and direct satellite signal (DS). The equations for PSD of a reflected signal from target, power of reflected signal, receiver input and maximum detection range were presented by researchers. In a calculated example the maximum detection range was 30Km for a disadvantageous scenario. Two types of antennas and a helicopter as a target were used in this experimental setup. The researchers concluded that detection of signal in the context of DS is one of the most difficult problems in the proposed system design. Researchers identified a number of approaches for DS signal suppression, including wave splitting, frequency and spatial adaptive filtering, and polarization.

A research was performed by Fabrizio Santi and Mikhail Cherniakov in which satellite navigation systems were studied as opportunity transmitter for both bistatic and multistatic SAR's [59]. Because many satellites with varying viewing angles are available in the area of interest, a multi-static SAR can be acquired using a single receiver at or above the ground surface. Signals from numerous satellites and reflected from the area of interest can be collected by a receiver on a stationary or moving platform. This research considers the fixed ground based receiver that is able to receive signals from several satellites and thus it can collect number of different bistatic images of the targeted area. The use of different spatially spaced satellites enabled a multiple perspective analysis of the area, which consisted of identifying the major aspects of the scene from each individual image and then integrating the information, thereby increasing the image's information space. In this research the features of image were identified by a multistatic image that was formed by combining single images at image level. For this purpose, To analyze GNSS-based SAR imagery and remove dominant scattering centers from the images, a method known as the ad-hoc clean algorithm was designed. A fluctuation in the position of the major scattering centers was to be expected when studying the scenario of widely separated satellites; With a CLEAN bistatic, the

scattered centers were extracted from the individual images and the information was then combined to form a scatter map. The effectiveness of this approach has been demonstrated through simulation and experimentation to determine the general structure of area [59].

While considering the case of limited variations in the perspective of the satellite, It is feasible to create a multistatic image with prominent scattering centers that are quite close in position to those found in single bistatic images. For this case multistatic image was used to extract the point features. According to authors, the images obtained from different satellites at different locations has differently oriented image responses, so the GNSS-based MSAR image can be distinguished by significantly improved PSF compared to BSAR images. In that way, a better performance can be obtained as compared to bistatic case and more information can be extracted from the scene. An enhanced feature extraction was obtained while considering the case of close scattering. These two qualities have also been demonstrated in simulation and experimental based images and they demonstrated the effectiveness of multistatic focusing in extracting not only the general characteristics of scenes, but also details. According to the authors, the technologies developed are not confined to passive GNSS-based systems, but can also be employed in principle in all MSARs [59].

3.6 Image Fusion

It is stated in [60] that the result of the two images is a combined image and basically an enhanced version of the two original images that contain all the basic information. Many real time applications required image fusion such as: surveillance applications, remote sensing applications, photography applications and medical applications. This study mentions image fusion methods such as Averaging, Brovey Method, Principal Component Analysis, Intensity Hue Saturation, Select Maximum Method, and Select Minimum Method. Pixel fusion of non-

transform based and sparse representations, Multi scale representations, and Hybridization of numerous transforms are the major approaches to fusion methods mentioned in this study. According to the experts, combining multiple fusion algorithms is particularly effective in the field of picture fusion. But there are some difficulties such as high-quality imagery equipment, displaced objects, complex computational algorithms, and combining characteristics such as directivity, multiple resolution, image noise, and inaccurate blurred results by image fusion [60].

In 2001, a research performed by G. Simone and A. Farina reveals a methodology that was helpful in gathering the data collected from different sensors in a single image and then can be used to extract more information from it [61]. The principle behind this research was based upon acquisition and processing of images of the same area obtained by different sensors at different time instants and then combined those using neural networks to extract information about the same area for different times. The sensors that acquired images were classified to as aerial or space sensors placed in series on an aircraft or satellites, they can also obtain information in different spectral bands based on the operating frequency or with different resolution. Therefore, there may be a wide range of data available for the same site and this data provided by individual sensors may not be sufficient for many remote sensing applications. Fusion of different images can help to gather information and provide better understanding of the area. In this article, the authors have successfully introduced SAR interferometry as an application in which the combined use of images obtained from two different antennas allowed the creation of an elevation map of the observed area [61].

Another research performed by Krista Amolins and Yun Zhang depicted a way to fuse images by using wavelet based technique. The authors have indicated that traditional imaging methods typically manage to inject multi-spectrum images into spatial detail, but often distort color-based information. The authors compared various wavelet-based methods in the basic,

hybrid, and advanced diagram categories. They observed that their proposed approaches were well carried out in comparison with other ways and had clear advantages over other methods, such as: rapid processing, reduced difficulty and memory needs, but most of the methods were tested on spot images, except for some that were tested on IKONOS, QuickBird or Landsat images. It is not possible to draw conclusion about which scheme works better until it is applied on all the above mentioned methods. The authors concluded that the wavelet based method performs better than the standard methods especially in terms of minimizing the color distortions and the results could be improved more to get more detailed information but that kind of schemes often have greater setup requirements [62].

4 SIMULATIONS

During the simulation process, a simple code was developed to evaluate imagery ideas and theories during the initial stage of the simulation. The code was modified after successful initial simulations to facilitate imaging in a viable scenario and efforts have been made to increase calculating efficiency.

It began with generating and assigning the individual PRN code for any GPS-satellite, as illustrated in figure (4.1), the characteristics of these codes during their transmission and their effects on the way like the delay and offset introduced, and finally the reception through a GPS receiver containing dual front end. By using the matched filter concept, a grey scale image was constructed. During the initial stages of simulation, this was not simulated because the up and down conversions are canceled during the modulation and demodulation process.

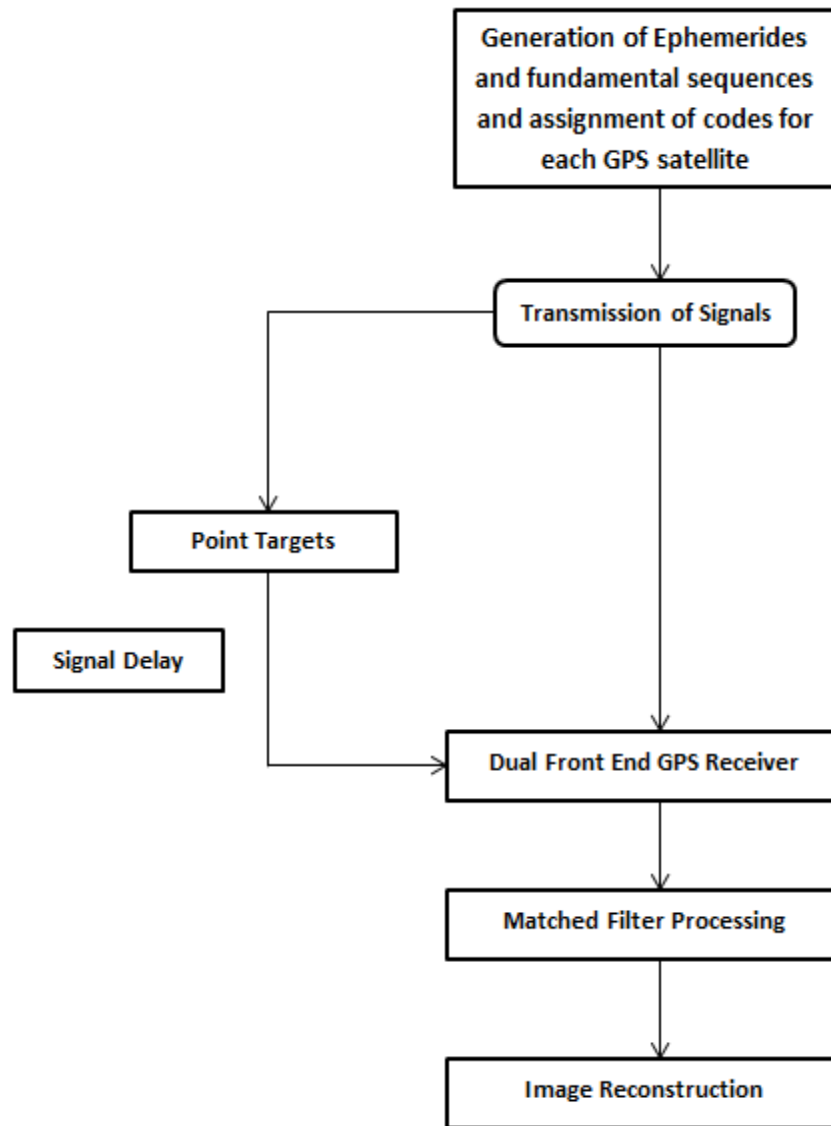


Figure 4. 1 Block diagram of the simulation code for imaging system

The simulations can be separated into two categories, firstly a real-world scenario simulation that involves the process in the space sector of GPS and space transit, orbits and the satellite coordinates, generating code, noise simulation and mitigation system. This includes a simulation process. The second is the simulation of the GPS receiver, including calculations of code offset and delay, correlation, processing of matching filter and processing of images.

The code was later modified to make it more realistic and adaptable to practical applications. The simulation has been divided into two different parts: the first part is in charge of

synthesizing the signal emitted by the analog-digital converter for the front-end of the GPS, and the second part is in charge of the reconstruction of the image. It has been taken care of that digital GPS signal and "ground data" such as satellite ephemeris records, location of the receiver and velocity are only information available to the reconstruction engine. The simulated signal was replaced by real GPS data after the reconstruction techniques were enhanced and finished.

The signal synthesizer has determined the satellite orbits and the individual change in the PRN code and integrated the direct GPS satellite signal with the reflected signals from all potential positions with the receiver noise. The transit times are determined based on the satellite and the receiver paths connected with the signal routes. The further attenuation was based on the hypothesis of points that were subjected to reflected signals.

4.1 DETAILED DESCRIPTION OF CODE

The program starts by defining globally specified variables, time, noise, pixel size and search region. In the first place, the area in which the object can be placed is 2000×2000 meters. The position fixed in the middle of search area is (0, 0, 6378050) which are the coordinates of the North Pole that are employed with a purpose of simplifying the coordinate transformation. The array of candidate position is multidimensional having 3D, each of the one for x, y, z coordinates. Initially, two targets were defined at positions (20, 0, 6378050) and (-20, 0, 6378050) respectively. Initially the pixel size was 20 meter and numbers of targets were modified later. The receiver position and velocity were set at (0, 0, 6378050) and (0, 0, 0) m/s respectively. The GPS satellites that have been selected are 1, 7, 9, 14, 16, 22, 25, 26 and 27. Any of these space vehicles (SV) can be allocated as the reference satellite. Targets are assumed to always reflect the signal with full power to the receiver.

4.1.1 Code generation

Table 4. 1 Code phase assigning and initial C/A code sequences

SV PRN Number	C/A Code Tap Selection	C/A Code Delay Chips	First 10 C/A Chips
1	$2 \oplus 6$	5	1100100000
2	$3 \oplus 7$	6	1110010000
3	$4 \oplus 8$	7	1111001000
4	$5 \oplus 9$	8	1111100100
5	$1 \oplus 9$	17	1001011011
6	$2 \oplus 10$	18	1100101101
7	$1 \oplus 8$	139	1001011001
8	$2 \oplus 9$	140	1100101100
9	$3 \oplus 10$	141	1110010110
10	$2 \oplus 3$	251	1101000100
11	$3 \oplus 4$	252	1110100010
12	$5 \oplus 6$	254	1111101000
13	$6 \oplus 7$	255	1111110100
14	$7 \oplus 8$	256	1111111010
15	$8 \oplus 9$	257	1111111101
16	$9 \oplus 10$	258	1111111110
17	$1 \oplus 4$	469	1001101110
18	$2 \oplus 5$	470	1100110111
19	$2 \oplus 6$	471	1110011011
20	$4 \oplus 7$	472	1111001101

21	$5 \oplus 8$	473	1111110011
22	$6 \oplus 9$	474	1111110011
23	$1 \oplus 3$	509	1000110110
24	$4 \oplus 6$	512	1111000011
25	$5 \oplus 7$	513	1111100011
26	$6 \oplus 8$	514	1111110001
27	$7 \oplus 9$	515	1111111000
28	$8 \oplus 10$	516	1111111100
29	$1 \oplus 6$	859	1001010111
30	$2 \oplus 7$	860	1100101011
31	$3 \oplus 8$	861	1110010101
32	$4 \oplus 9$	862	1111001010
33	$5 \oplus 10$	863	1111100101
34	$4 \oplus 10$	950	1111001011
35	$1 \oplus 7$	947	1001011100
36	$2 \oplus 8$	948	1100101110
37	$4 \oplus 10$	950	1111001011

The PRN sequences are generated that have separate codes for each satellite and then assigned to an individual satellite vehicle (SV). Code generation in both types of simulations is performed first by the GPS satellite and then by the receiver. To perform the correlation process, receiver has to reproduce the code. As it is explained in the previous chapter, the generation of C/A code involves two 10 bits shift registers that generate PRN sequence with a length of 1,023 bits. G1 is the first shift register with a feedback tap linked to stage 3 and 10 which is then returned to stage 1. The delay of the G2i sequence is constructed by an integer chip number ranging from 5 to 950. Two G2 taps are linked with each PRN C/A code number. The tap combination of all defined GPS PRN's is described in table (4.1) and equivalent delay is also specified in the C/A code chips. The space segment reserves first 32

PRN's and five PRN's (33-37) are reserved by ground transmitters including user, while 34 and 37 are identical.

4.1.2 Calculation of code offset

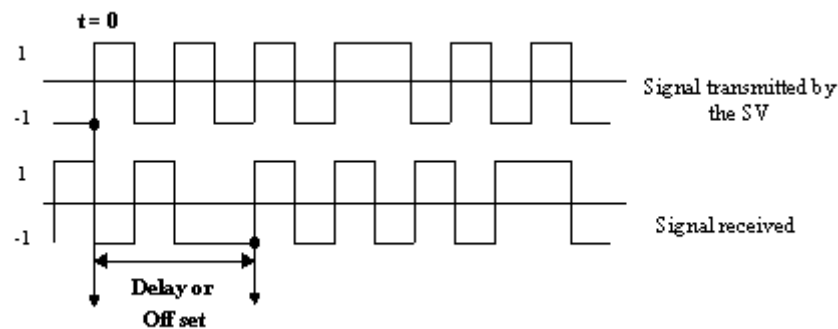


Figure 4. 2 Code delay

Each GPS satellite generates a specific PRN sequence, using the commands in Matlab, which begins at $t = 0$ for each satellite. In a practical scenario, however, the signals on space require some (very small) time to reach the receiver. We can employ SV to simulate the delay so that we don't need to calculate the total delay, simply measure the delay difference.

Matlab now includes a function for calculating the delay difference, which delivers the delayed PRN version. This function makes use of the PRN sequences of all GPS satellites, the chip rate, the weekday difference, and the delay. The difference is referred to as the offset, and the PRN created is referred to as offset PRN. This function calculates the offset of the direct and reflected signals. In phase (I) and quarantine phase (Q) components, this offset is employed to define the GPS signal base band in accordance with the following sections.

4.1.3 Display of image

Figure (4.4) depicts the search region where targets can be found. To model the area in which the target could be positioned, two arrays of size 101 each are chosen, and the pixel size of the target is 20 meters, resulting in a $20 \times 100 = 2000$ square meter area. In the matrix, the size of the target is referred to as an element, pixel, or square, and the size of 20 meters was chosen at random in this initial simulation.

To improve image resolution, the target grid was resized later in the experiment. However, the array and target sizes can be lowered to improve computing efficiency. The target candidate position can be chosen from any location in the matrix. The search area should be centered at the position $(0, 0, 6378050)$. In the ECEF coordinate system, these are the coordinates found in the center of the North Pole and are used to facilitate the coordinate conversion.

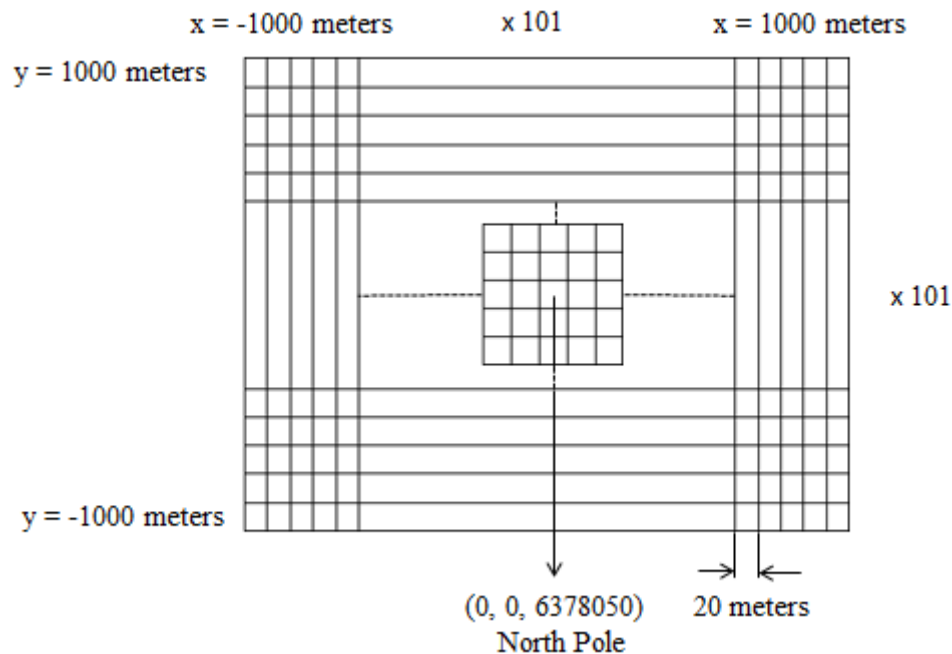


Figure 4. 3 Candidate position for target

4.1.4 Image reconstruction and matched filter processing

In signal processing, an approach based on matching filters is used to improve system resolution. This is accomplished by bit-by-bit multiplication of two codes and then adding them in Matlab. For each sample or period, the process is repeated, and the associated value is stored in a 101 x 101 matrix, as illustrated in Figure (3.4). The matrix is scanned over time to match the matching filter process for each sample. In actuality, the code multiplies the signal and the delayed version and then adds it for all periods and this is almost identical to sample integration. The location of the object will be high correlation or of a maximum value.

The following equation converts the values of the baseband components in phase (I) and quadrature phase (Q) into comparable magnitude:

$$\text{Equivalent Magnitude} = \sqrt{I^2 + Q^2} \quad (4.1)$$

The grey scale image is created using the information provided above. The colors of the grey scale image are a combination of black, white, and grey tones. In the above-mentioned simulation, the Matlab function returns the maximum value; the data is inverted and adjusted so that one is the maximum value. When there is highest correlation, a black pixel or square will exhibit the data and, if there are little correlations, a white pixel or a square will display the data, so we do not grasp the object. This could also be the opposite case. As a result, the generated image directs the target to the appropriate coordinate of the candidate's position.

4.1.5 Simplifications and assumptions

Different assumptions and simplifications have been made to facilitate simulation. Point target theory of resolution is applied and it is assumed that the target is a fixed point scatterer. Signal degradation due to RCS is not considered. Furthermore, all signals are considered to

be reflected back to the receiver by the target, and the reflection coefficient is assumed to be the same.

4.2 IMPROVEMENTS AND MODIFICATIONS IN THE CODE

In order to realize the objective to build up a viable imagery system, actual data samples from the GPS antenna must be used via the RF front end and the ADC. However, it is not desirable for real data samples to be used in the initial stage of signal processing algorithm development. Because controlling of the received signal is not only impossible, it is also difficult to know all of the features of the received signals. The Matlab coding that produces a GPS-like signal has been created with the IF(few MHz) specified and other parameters, such as noise and attenuation. With the routine Matlab approaches the signal might be uploaded in a file. The reconstruction engine was additionally modified and detailed. A signal generator that can generate a GPS signal with required characteristics is shown in figure (4.5).

Form1

Mixers

Reference Crystal

Nom. freq: 2 MHz

Error: 0 ppm

Xtal freq: 2 MHz

Ref Divisor: 10

LO1 Multiplier: 8,064

LO2 Divisor: 48

Use Internal SampleClock ☐

Ref freq:	200	KHz
LO1 freq:	1612.8	MHz
IF1 freq	37.38	MHz
LO2 freq:	33.6	MHz
IF2 freq	3.78	MHz
Sample Clock:	2	MHz

Run Details

Duration: 1 sec

Rx Antenna Gain: 30 dBi

Direct Suppression: 0 dB

Noise Enabled: ☐

PR0=PointReflector

Go!

Figure 4. 4 Screenshot for main form of signal generator

4.2.1 Resolution

The area simulated was a 400 x 400 pixel matrix of 25 x 25 cm to test the best feasible resolution. Two point reflector targets have been placed in the area of interest having coordinates of (20, 0, 6378050) and (-20, 0, 6378050). Each conductive point reflector was designed in the shape of a sphere and had an effective radius of 10 cm and a reflection coefficient of 1. This particular radius value was chosen to show that a target comparable to the 20 cm wavelength for GPS L1 frequency can be detected. The reflected signal antenna

was expected to have an increase of 30 dB and to suppress the direct signal by 70 dB. The coherent integration time has been set at 100 s, which is followed by 10 s net integration time, but with the code segmentation as mentioned earlier.

4.2.2 Image generation

The strength of the reflected signal is based on surface roughness, dielectric qualities of the reflective objects as well as the number of reflections as indicated earlier. In accordance with the inverse square law, a rate of around $20 \log_{10} (R/0.1)$, whereby range of reflection point is denoted by 'R' to the observer' is attenuated for the signal reflected in a conductive sphere of a radius of 10 cm. For instance, the attenuation of the RHCP antenna across a range of 1000 meters would be around 80 dB lower than direct signals. The SNR will be roughly -96dB for a 2MHz bandwidth receiver. Due to an antenna gain (30 dB) this is enhanced to -66 dB. With an integration rate of 2 MS/s, we have 2×10^8 samples in the time of 100 s, an improvement of 83 dB, giving the reconstructed image a predicted PSNR of 17 dB.

While the object display a star-forming PSF, achieving a comparable wavelength of the GPS L1 frequency is a new accomplishment, using GPS reflected signals and without the assistance of a dedicated transmitter. No part in the research literature has been documented, to the best of the author's knowledge. When choosing a certain type of radar for various purposes, resolution is a crucial consideration. The increased resolution will undoubtedly pave the way for new applications for these imaging devices. This is a substantial improvement in the realm of remote sensing by utilizing opportunity transmitters, particularly GPS.

4.3 Imaging Scenarios

There could be three kind of different imaging scenarios as mentioned in the initial chapter. This depends upon positioning of the data acquisition hardware or GPS receiver and source of synthetic aperture. The hardware could be airborne, mounted on a slow moving ground based vehicle or could be static. The simulations regarding a static receiver have been explained in this thesis.

4.3.1 Static receiver

The concept to generate an image through the correlation of direct GPS signals is founded on the SAR principle, which requires the receiver or bistatic radar to maintain airborne or spatial mobility. However, if the sampling is carried out over a longer period, it is conceivable to construct an image with a stationary receiver, at least in theory. The image is expected to reveal important terrain features with acceptable spatial resolution. Since the receiver is fixed, the change in geometry is provided by the GPS satellite in motion, as shown in Figure (4.7).

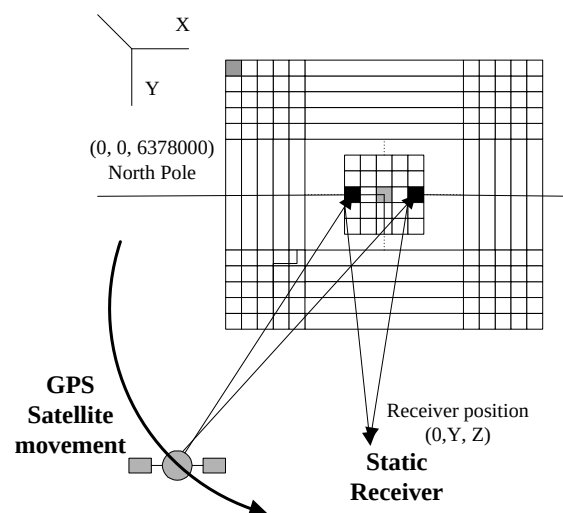


Figure 4. 5 Imaging scenario for static receiver

It has been mentioned in many research papers that this technique can be used for imaging purposes. In fact, several institutions are studying BSAR with an asymmetric configuration (some of them were mentioned in literature review chapter) however, there is limited information on thorough simulations to confirm technical viability or experimental outcomes. Efforts to simulate and validate results in reality with a static GPS receiver are a novel achievement in this domain.

The motivation to perform image generation with the antenna and data acquisition hardware remaining static originated from the fact that it is difficult to arrange a moving platform due to the associated logistic issues. Moreover, certain application like long term seismic and terrain monitoring applications require the receiver to remain static, thus enabling it to monitor certain landscapes constantly, day and night and in any weather conditions.

4.3.1.1 Optimum integration time for static receiver data acquisition

Average orbital altitude in GPS satellites is 20,200 kilometers above the surface of Earth, with one orbit taking approximately 12 hours to complete. [2]. As a result, the satellite can be seen over the horizon for six hours. In practice, obtaining GPS data consistently for more than two to three hours is challenging due to the fact that unobstructed access can be limited by neighboring trees, foliage, and buildings. As a result, the practical image generation time for imaging with a stationary receiver is roughly 2 hours (7200 seconds), given that the reflected signal is available during this time and that the receiver and antennas are properly positioned to receive both the direct and reflected signals.

It is extremely difficult, if not impossible, to obtain two to three hours of continuous data at a sampling rate of 19.2 MHz. The size of data may become unmanageable as two hours of

continues data, at this sampling frequency, requires 138Gbytes of storage space and with 500GB of storage disks available commercially the image formation may be limited to a couple of applications. Thus, continuous data acquisition and monitoring and need for documenting the data and images will render such huge data storage impossible to handle.

The main issue lies in the speed of the Matlab reconstruct code which takes 10 to two hours to reconstruct 10 seconds of data and approximately 12 hours per 100 seconds of data to enhance image quality and clarity (depending on the area of interest). As a result, the time required for processing 2 hours (7200 seconds) of data is extrapolated; the generation of image could take several days. This could make the system useless for practical applications. The reconstruction engine is modified and uses segmentation techniques due to which it is expected to take care of both issues.

4.4 Modification of image reconstruction algorithm

Detailed simulations support the idea that a high-resolution image can be formed by utilizing the correlation features of GPS signals and satellite movement to construct a synthetic aperture. However, in reality, the imaging task can be difficult due to the poor signal-to-noise ratio of a weakly reflected GPS signal, the availability of GPS signals for an extended length of time, hardware restrictions, and other issues.

5 RESULTS

5.1 INITIAL SIMULATIONS

As mentioned in the last chapter an initial and comparatively simple code was compiled to verify the imaging concepts. The simulations were carried out in Matlab under different scenarios and conditions as well as by varying some of the parameters as shown in table (5.1).

Table 5. 1 Parameters for simulations

Description	Values
Integration time (Ts)	0.1 seconds
Attenuation factor	0.5
Target No 1 coordinates	(20, 0, 6378050)
Target No 2 coordinates	(-20, 0, 6378050)
Receiver coordinates	(0, 0, 6378050)
Receiver velocity (m/s)	(0, 0, 0)
GPS satellites involved (PRN)	1, 7, 9, 14, 16, 22, 25, 26, 27

5.1.1 Initial results

The simulation was carried out with an integration time of 0.1s, with other values as mentioned in Table (5.1), and results the figure (5.1). Both targets are easily recognizable with acceptable spatial resolution. During initial simulations, targets were identified as a pixel (20 m) or element at the candidate position. A simulation was also performed at an integration time of 0.01s which consumed very less time, but the targets were not resolved at cost of low resolution and introduction of the noise.

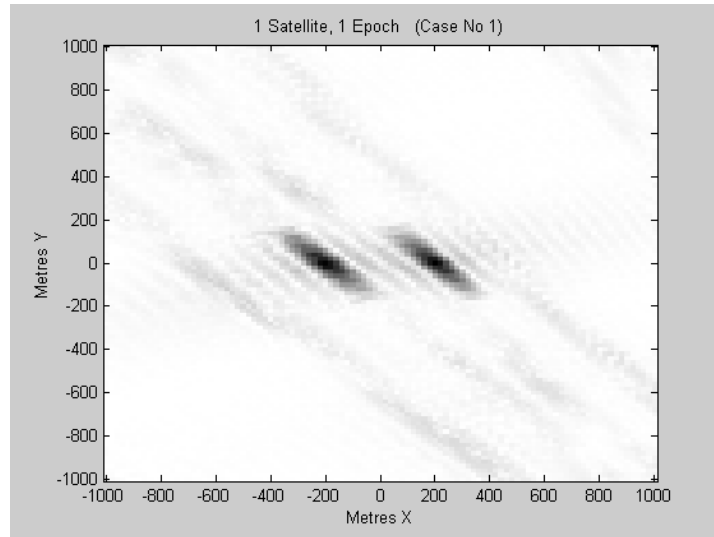


Figure 5. 1 Integration time of 0.1 second

During the initial simulations, the following interesting results were observed:

1. In reality, utilizing matching filter processing based on SAR techniques; the image can be formed by correlating the direct and reflected signals from GPS. Increasing observation time results in more samples, resulting in better simulation results and higher resolution by canceling unrelated noise

It was concluded by the initial simulations that the Matlab compiler algorithm and programs are proved to be expensive in computational time and memory. The code took approximately 25 minutes to run in a Matlab environment on a laptop with 2.10 GHz processor while the integration time was kept to 0.1 second and about 45 minutes when the integration time was 0.2 seconds.

5.1.2 Static receiver

As previously stated, this type of system's setup can contain a stable GPS receiver. If samples are taken every few seconds for a longer length of time, say 6 hours or more after integration,

the software should produce comparable findings for the stationary receiver. The software was modified and Simulations showed that for long durations the targets can be resolved via integration, while significantly reducing the rate of over-sampling.

5.1.3 Effect of integration time

The integration time was kept at 21600 seconds (equal to six hours). The results depicted that the resolution has improved considerably. A look at figure (5.2) disclose that both the targets were with perfect resolution visible as one distinct pixel each and devoid of any distortion. As stated earlier, an increase in integration time improves the SNR by cancellation of uncorrelated noise and better resolution on account of the increase in synthetic aperture. However, in case of practical data acquisition and imaging, the requirement of longer integration time for a static receiver is expected to cause unique challenges. It will be very difficult to sustain the acquisition of GPS signals for multiple hours due to lack of signal availability and poor satellite geometry. The loss of signal lock will result in excessive noise and compromised resolution.

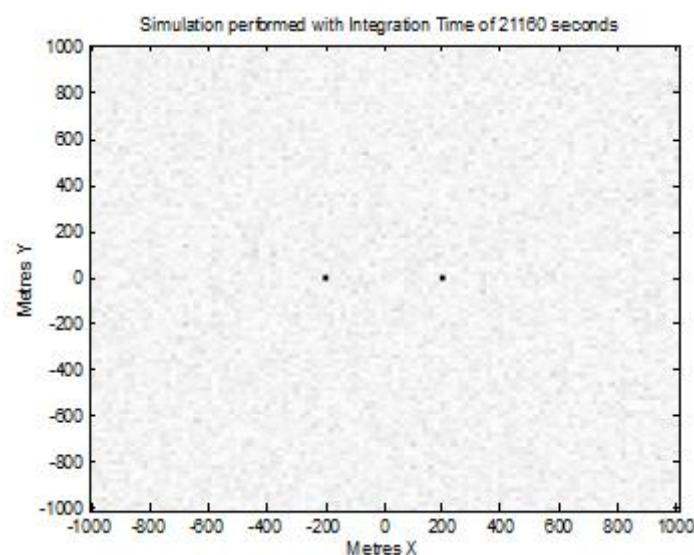


Figure 5. 2 Integration time of 21600 seconds

It is imperative to consider that these initial simulations were performed with many simplifications and assumptions. The main aim of these simulations were only to verify the imaging concept.

Figure (5.2) shows how this technology can be utilized in terrain monitoring applications to generate detailed and cost-effective photographs of a specific site. However, it is acknowledged that the resulting image may be badly damaged in a practical context. Additional research will improve the image and make it as close as possible to the ideal. Some factors that adversely affect the image include the strength of reflected signals, noise atmospheric interference, radar clutter, and hardware limitations.

The above simulations have been performed with the simple code simulating a static GPS receiver. However, it is expected that the results and conclusions regarding threshold of target detection, simulation of atmospheric effects, simulations of receiver clock errors and accuracy of the ADC are also applicable for other imaging scenarios and arrangements.

5.1.4 Threshold of target detection

As mentioned earlier, the strength of reflected GPS signal is expected to be much lower and submersed in excessive noise. Simulations were performed regarding the threshold at which the reflected signal will be extremely weak to resolve the targets and have been documented in table (5.2).

Table 5. 2 Results comparison for observing the detection threshold of reflected signals

Description of simulation	Results observed	Conclusion
---------------------------	------------------	------------

parameter		
With direct signal at 0.001 (as compared to initial value of one), the indirect signal was reduced from 0.9, 0.5, 0.25, 0.2, 0.1, 0.01 and 0.001.	The targets were resolved with all these values of indirect signal.	The threshold for detection of targets (in the absence of noise) was when the reflected signal is at
The indirect signal was further reduced to 0.0005 (-26dB), 0.000025 (-32 dB) and 0.00001 (-40 dB) of the original value.	From the results it was clear that noise has started to appear in the background when the reflected signal was at 0.0001, at 0.00005 the target were still visible, but there was no target present when the reflected signal was 0.00001 of its original value.	0.000025 (-32 dB) and direct signal at 0.001 of its original value.

In comparison with the maximum correlation of the direct signals, the signal that is reflected should not be less than 24 dB. The results highlight another finding that contrary to the theoretical, the threshold for detection of targets (in the absence of noise) is 32 dB. Thus the dynamic range for the detection of reflected signals is wider as compared to the theoretical limit. The results can be observed in figure (5.3).

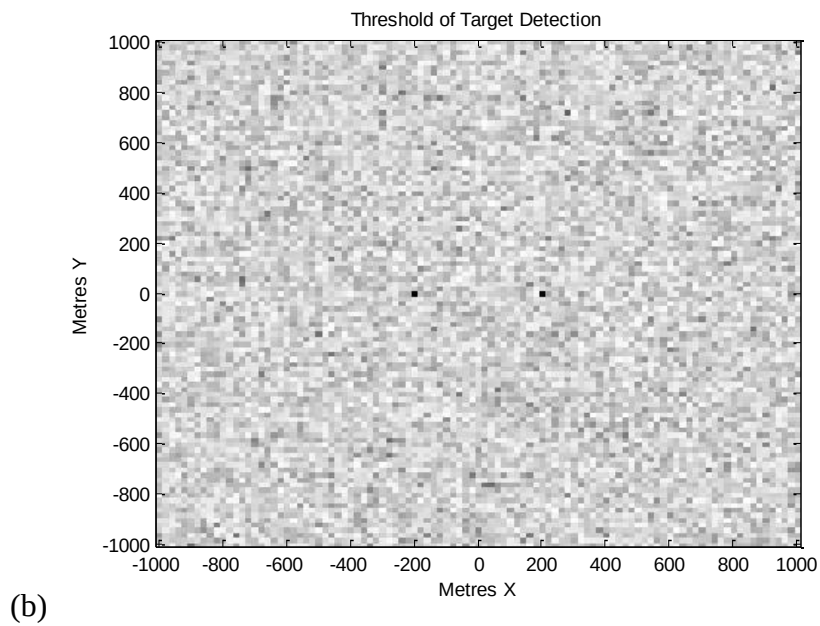
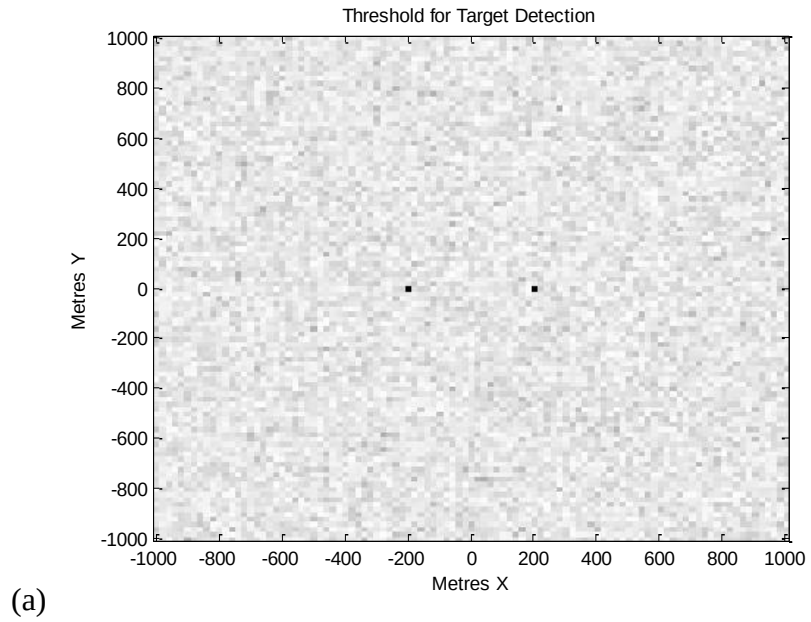


Figure 5. 3 Comparison of results for (a) reflected signal at 0.00005 and direct signal at 0.001 (-26 dB) (b) Reflected signal at 0.000025 and direct signal at 0.001 (-32 dB)

The average Position Dilution of Precision (PDOP) is 7.8783, as shown in the sky plot in figure (5.11). When satellites are packed together, high PDOP values are discovered. Experimentation has revealed that it is ideal if the satellites are evenly placed around the

receiver, with one on top and one down on the horizon. When the PDOP is less than 5, good observations have been achieved with measurements of not less than 5 satellites [24].

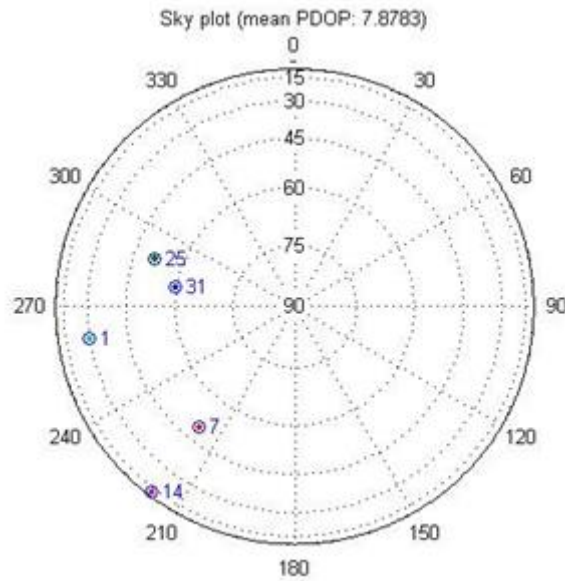


Figure 5. 4 Sky plot, showing the visible satellites

5.2 SIMULATION RESULTS FOR DIFFERENT SATELLITE GEOMETRIES

As previously stated in this thesis, a stationary receiver can cause a change in the geometry of a revolving GPS satellite. During the simulations, it is discovered that the ideal integration time for a reception speed of 200 m / s is 10 s, which is a compromise between computing efficiency and resolution and data volume. During this period, the receiver travels a distance of 2 kilometers, and picture reconstruction takes only a few minutes for tiny arrays of targets. To determine the optimal time required to reconstruct an image in the event of a stationary receiver, a different signal was generated using a GPS signal simulation code, with a step-by-step decline in receiver speed and increased integration time (T_s). The net integration time (T_n) was fixed at 10 s to ensure that the targets were resolved. This was achieved through segmentation introduced into the reconstruction algorithms to process longer duration signals.

Some other parameters that were changed are shown in the table (5.3).

Table 5. 3 Optimum time for static receiver simulations

Total integration time (T_s , s)	Net integration time (T_n , s)	Receiver velocity (R_v , m/s)	Number of segments (SS)	Skip time (T_{sk} , s) (per segment)
10	10	(200, 0, 0)	1	0
100	10	(20, 0, 0)	10	9
200	10	(10, 0, 0)	20	9
400	10	(5, 0, 0)	40	9
800	10	(2, 0, 0)	80	9
1600	10	(1, 0, 0)	160	9
3200	10	(0, 0, 0)	320	9
3600	10	(0, 0, 0)	1000	3.59
7200	10	(0, 0, 0)	2000	3.5950

As shown in table (5.3), the targets appear to be solved with a fixed receiver at first, with an integration time of 3200 seconds and a count of 320 segments. An enhanced image was revealed by increasing the number of segments to 1000. The number of segments were increased to 2000 and the integration time (T_s) was increased to 7200, resulting in improved image quality and resolution.

As a result of the simulations, at least 1 hour (3600 s) of integration time is required to successfully resolve the targets with an adequate spatial resolution. 2 hours of data, on the other hand, delivered the sub-meter level range resolution. Another important finding is that the data should be segmented as much as possible. By correlation, it is crucial to attain a weak reflected signal and so cancel unlinked noise. T_{dw} must be chosen with caution because the increased dwell time can reduce computing efficiency and result in ground clutter

obscuring the targets. In contrast, a low Tdw will not result in the collection of a weakly reflected GPS signal.

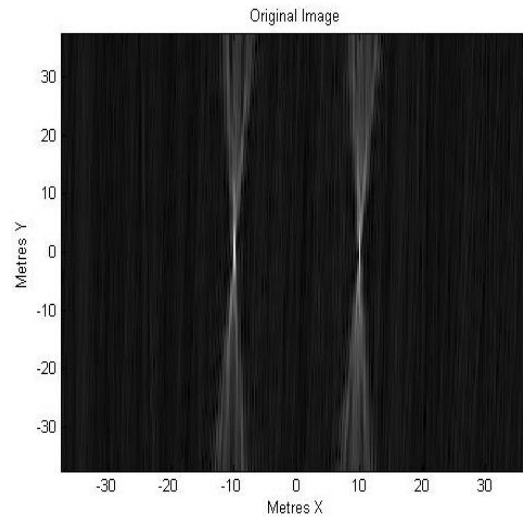


Figure 5. 5 Simulation results for 7000 seconds

The result obtained by an integrating time of 7000 s is shown in Fig. (5.12). Two targets of 10 cm appeared to be resolved with sub meter range resolution. It has already been shown that resolution is directly proportional to the change in the geometry of the GPS satellite in orbit. It will be fascinating to examine how the total time of integration impacts the quality and resolution of images. More synthetic openings are generated by increasing integration time.

5.2.1 Effect of satellite geometry

The location of GPS satellite with respect to the receiver can have a profound effect on the image and resolution. Figure (5.17) compares the reconstructed image for a simulated signal of length 7200 seconds and net integration time of 10 seconds using different GPS satellites as transmitters of opportunity. GPS satellite with PRN 27, 16 and 25 have been selected.

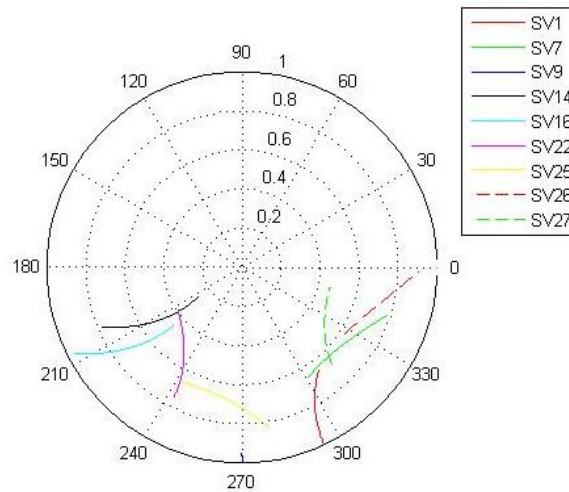


Figure 5. 6 Polar plot showing location of satellites for 7200 seconds

It is also essential that during the acquisition period, the geometry of GPS satellite should be optimal. Furthermore, it is better to have satellites orbiting from north to south direction during data acquisition for improved range resolution, but practically it is not always possible.

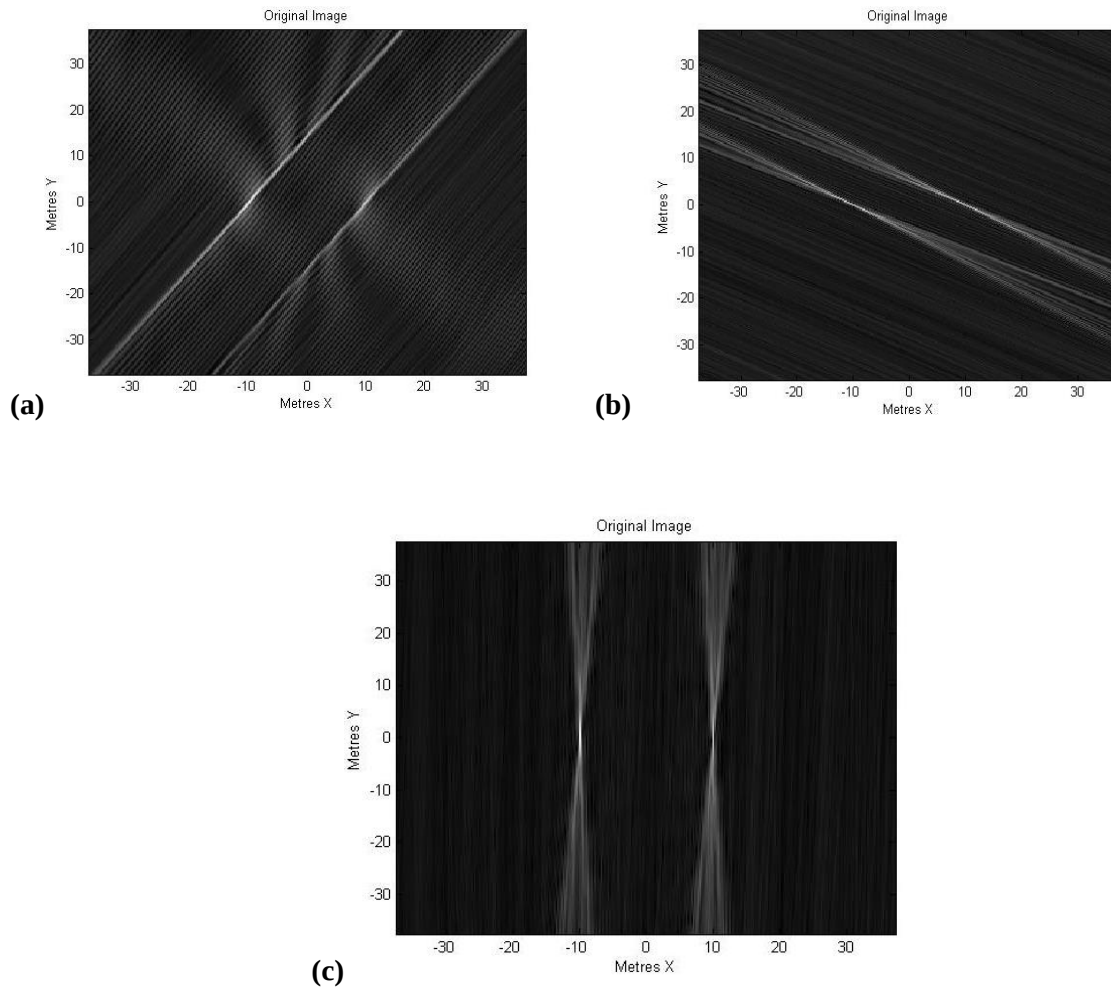


Figure 5. 7 Comparison of results for images obtained for an overall integration time of 7200 sec and net integration time of 10 sec for GPS satellite (a) SV 16 (b) SV 25 (c) SV 27

5.3 IMAGE FUSION FOR SIMULATED IMAGES

The images obtained from different satellite vehicles are fused with different combinations with each other which resulted in pin pointing the target locations in each case. All fusion results are giving approximately same location for both targets which declare this method to be more useful.

A combination of image fusion for SV 16 and SV 25 is shown in fig. (5.18). The image fusion of SV 16 and SV 27 is shown in fig. (5.19) and image fusion for SV 25 and SV 27 is shown in fig. (5.20).

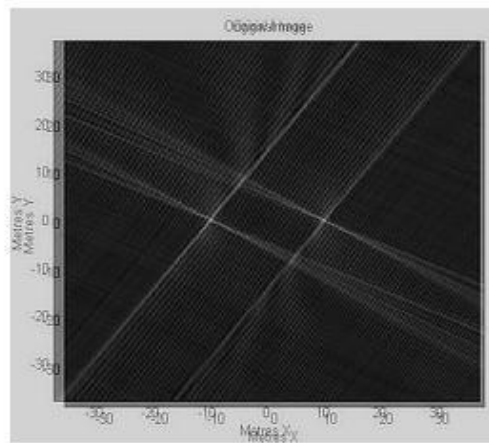


Figure 5. 8 Image fusion for SV 16 and SV 25

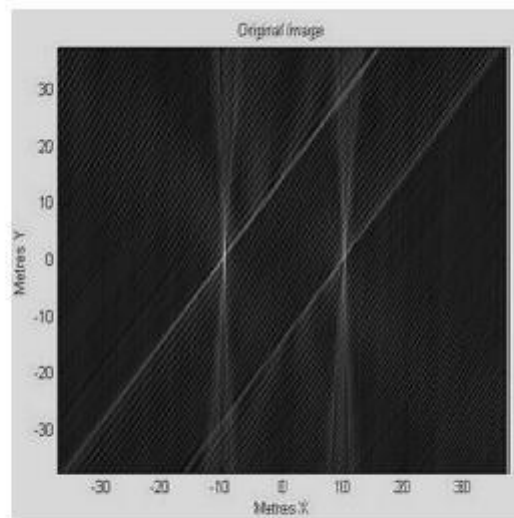


Figure 5. 9 Image fusion for SV 16 and SV 27

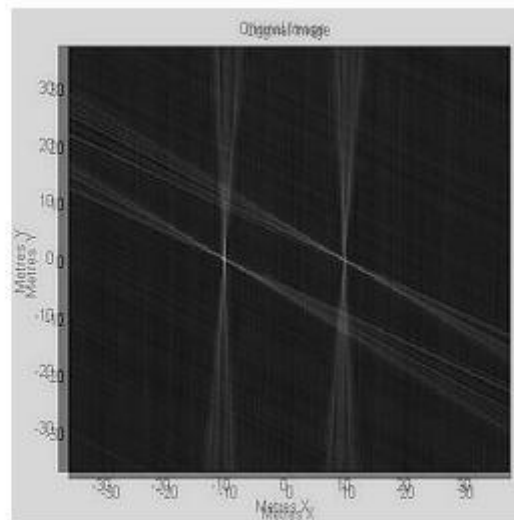


Figure 5. 10 Image fusion for SV 25 and SV 27

FUTURE WORK

Although a steady progress has been achieved and an image of the area of interest has been created, with a target of around 0.5 meters. There are some developments and issues that can be addressed further to improve the imaging system.

Some extensive field experimentations need to be performed with imaging device. In order to verify the target detection, further data must be collect at a suitable position to ensure that the reference GPS satellite is visible for a high resolution image for at least two to three hours.

The possibility of receiving more GPS satellites is to be done by a signal acquisition in open environments to provide clear views of the sky.

A reasonable solution is a new accomplishment to obtain an image of the area of interest and no research group or individual with this specific method has simulated or shown an imaging system. However the resolution can be improved more by clutter rejection, increasing number of segments, by investigating reference coordinate transformations and trying other image fusion techniques.

For low-cost and detailed information on the target location, the proposed method can be employed on high-resolution terrain and monitoring applications. However, the resulting image can be drastically damaged in an actual environment. The noise, reflective strength, the disguise and the interference with direct signal, atmospheric impacts, radar confusion and hardware restrictions are some elements which have negative effects on the signal.

CONCLUSION

The objective of this research was to simulate and obtain high resolution images of the area of interest. During this research the imaging arrangement was studied and simulated in a detailed manner. As mentioned in the first chapter, simulations have been performed for a static receiver in order to get images from different satellite geometries. Although the results have confirmed the objective of this research but still there are lot of efforts required (described in future recommendations chapter) in order to exploit the imaging system for proposed commercial applications.

As a number of institutes examine BSAR with asymmetric structure, the idea of applying this technology for imaging applications has been noted in various research papers (some of them

have been mentioned in chapter 3 of this thesis) but there is very less information of detailed simulations to experimentally confirm the results and feasibility of the system.

The imaging algorithms can be changed to suit this system, which improves detection range due to its improved signal intensity and resolution due to its higher bandwidth. Other systems will add new indications of opportunity; therefore the future of this research area is predicted to be highly fruitful.

6 References

- [1] P. Misra, P. E.-G. P. S. Signals, undefined Measurements, and undefined 2006, “Global Positioning System: signals, measurements and performance second edition,” *tobuk.co*, Accessed: Jan. 15, 2021. [Online]. Available: <https://tobuk.co/393.pdf>.
- [2] E. Kaplan and C. Hegarty, *Understanding GPS: principles and applications*. 2005.
- [3] M. Usman and D. W. Armitage, “Acquisition of reflected GPS signals for remote sensing applications,” in *ICAST 2008: Proceedings of 2nd International Conference on Advances in Space Technologies - Space in the Service of Mankind*, 2008, pp. 131–136, doi: 10.1109/ICAST.2008.4747700.
- [4] C. I. Pipitone, A. Maltese, G. I. Dardanelli, M. I. Lo Brutto, and G. I. La Loggia, “Monitoring Water Surface and Level of a Reservoir Using Different Remote Sensing Approaches and Comparison with Dam Displacements Evaluated via GNSS,” *mdpi.com*, doi: 10.3390/rs10010071.
- [5] M. Ha, J. Darrozes, M. Grippa, ... E. M.-, and undefined 2018, “GNSS-R for detection of soil moisture and effective height changes using SNR phase unwrapping method: study case of Dahra (Senegal),” *ui.adsabs.harvard.edu*, Accessed: Jan. 15, 2021. [Online]. Available: <https://ui.adsabs.harvard.edu/abs/2018EGUGA..2013483H/abstract>.
- [6] E. Valencia, ... A. C.-I. J. of, and undefined 2012, “Using GNSS-R imaging of the ocean surface for oil slick detection,” *ieeexplore.ieee.org*, Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/6293919/>.
- [7] M. Usman, D. A.-J. of G. P. Systems, and undefined 2009, “Details of an imaging system based on reflected GPS signals and utilizing SAR techniques,” *academia.edu*,

- Accessed: Jan. 15, 2021. [Online]. Available:
http://www.academia.edu/download/35048532/JoGPS_v8n1p87-99.pdf.
- [8] M. H.-I. J. O. H. A. S. S. (IOSR and undefined 2016, “Basic concept of GPS and its applications,” *jnp.ir*, Accessed: Jan. 15, 2021. [Online]. Available: <http://jnp.ir/en/wp-content/uploads/sites/9/2017/08/gps-new.pdf>.
 - [9] “GPS: The Global Positioning System.” <https://www.gps.gov/> (accessed Jan. 15, 2021).
 - [10] S. P.-S. Policy and undefined 1996, “The global positioning system: policy issues for an information technology,” *Elsevier*, Accessed: Jan. 15, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0265964696000288>.
 - [11] N. Wang, Y. Qin, B. Z.-2007 I. S. on, and undefined 2007, “Direct and rapid P code acquisition technique,” *ieeexplore.ieee.org*, Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/4392185/>.
 - [12] A. E.-A. E. Journal and undefined 2011, “Enhancing the accuracy of GPS point positioning by converting the single frequency data to dual frequency data,” *Elsevier*, Accessed: Jan. 15, 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S111001681100041X>.
 - [13] G. Wu, C. Chen, N. Yang, H. Hui, P. X.-C. I. Automation, and undefined 2019, “Design of differential GPS system based on BP neural network error correction for precision agriculture,” *Springer*, Accessed: Jan. 15, 2021. [Online]. Available: https://link.springer.com/chapter/10.1007/978-981-32-9050-1_49.
 - [14] C. Kee, B. P.-I. transactions on aerospace and, and undefined 1996, “Wide area differential GPS (WADGPS): Future navigation system,” *ieeexplore.ieee.org*,

- Accessed: Jan. 15, 2021. [Online]. Available:
<https://ieeexplore.ieee.org/abstract/document/489522/>.
- [15] G. Beyerle, M. Ramatschi, R. Galas, T. Schmidt, J. W.-G. solutions, and undefined 2009, “A data archive of GPS navigation messages,” *Springer*, Accessed: Jan. 15, 2021. [Online]. Available: <https://link.springer.com/content/pdf/10.1007/s10291-008-0095-y.pdf>.
- [16] “GPS.gov: New Civil Signals.”
<https://www.gps.gov/systems/gps/modernization/civilsignals/> (accessed Jan. 15, 2021).
- [17] S. Banupriya, S. Kumar, I. Venkatesh, and M. Sasikala, “A study on PRN Code Generation and Properties of C/A Code in GPS Based on MATLAB,” 2019, Accessed: Jan. 15, 2021. [Online]. Available:
<http://www.academia.edu/download/59805155/IRJET-V6I423720190620-98492-17r7z7.pdf>.
- [18] J. Malmström, “Robust Navigation with GPS/INS and Adaptive Beamforming,” 2003. Accessed: Jan. 15, 2021. [Online]. Available: <https://www.diva-portal.org/smash/record.jsf?pid=diva2:956118>.
- [19] E. Akim, D. T.-T. P. of the 17th International, and undefined 2003, “GPS errors statistical analysis for ground receiver measurements,” *kiam1.rssi.ru*, Accessed: Jan. 15, 2021. [Online]. Available:
<https://scholar.google.com/ftp://ftp.kiam1.rssi.ru/pub/gps/lib/publications/issfd17.pdf>.
- [20] S. Raghunath, B. Malleswari, K. S.- IJORCS, and undefined 2011, “Analysis of GPS errors during different times in a day,” *Citeseer*, Accessed: Jan. 15, 2021. [Online]. Available:
<http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.219.3969&rep=rep1&type=>

pdf.

- [21] E. E. Altshuler, “Tropospheric Range-Error Corrections for the Global Positioning System,” 1998. Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/668906/>.
- [22] T. Kos, I. Markezic, J. P.-P. ELMAR-2010, and undefined 2010, “Effects of multipath reception on GPS positioning performance,” *ieeexplore.ieee.org*, Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/5606130/>.
- [23] E. M. COPPS, “An Aspect of the Role of the Clock in a GPS Receiver,” *Navigation*, vol. 31, no. 3, pp. 233–242, 1984, doi: 10.1002/j.2161-4296.1984.tb00875.x.
- [24] P. MASSATT and K. RUDNICK, “Geometric Formulas for Dilution of Precision Calculations,” *Navigation*, vol. 37, no. 4, pp. 379–391, 1990, doi: 10.1002/j.2161-4296.1990.tb01563.x.
- [25] S. Y. Zheng, “SIGNAL ACQUISITION AND TRACKING FOR A SOFTWARE GPS RECEIVER,” 2005. Accessed: Jan. 15, 2021. [Online]. Available: <https://vtechworks.lib.vt.edu/handle/10919/31291>.
- [26] B. Sklar, *Digital communications: fundamentals and applications*. 2001.
- [27] F. E. . Nathanson, J. Reilly, and M. Cohen, “Radar design principles: signal processing and the environment,” 1999, Accessed: Jan. 15, 2021. [Online]. Available: <https://pdfs.semanticscholar.org/a6d8/065b2d664bc0709eafbd6aa705935279f629.pdf>.
- [28] P. Z. Peebles, *Radar Principles*. 1998.
- [29] E. F. Knott, J. F. Shaeffer, and M. T. Tuley, *Radar Cross Section*. Institution of Engineering and Technology, 2004.

- [30] M. Usman, D. Armitage, M. Usman, and D. W. Armitage, "A Remote Imaging System Based on Reflected GPS Signals," *ieeexplore.ieee.org*, 2006, doi: 10.1109/ICAST.2006.313822.
- [31] "Radar Resolution and Complex-image Analysis - August W. Rihaczek, Stephen J. Hershkowitz - Google Books."
https://books.google.com.pk/books/about/Radar_Resolution_and_Complex_image_Analy.html?id=-AdTAAAAMAAJ&redir_esc=y (accessed Jan. 15, 2021).
- [32] M. Cherniakov, *Bistatic Radar: Principles and Practice*. John Wiley and Sons, 2007.
- [33] "NASA Jet Propulsion Laboratory (JPL) - Space Mission and Science News, Videos and Images." <https://www.jpl.nasa.gov/> (accessed Jan. 15, 2021).
- [34] "Radar Handbook, Second Edition (eBook, 1990) [WorldCat.org]."
<https://www.worldcat.org/title/radar-handbook-second-edition/oclc/1084400902> (accessed Jan. 15, 2021).
- [35] M. C. Jackson, "The Geometry of Bistatic Radar Systems," *IEE Proc. F Commun. Radar Signal Process.*, vol. 133, no. 7, pp. 604–612, 1986, doi: 10.1049/ip-f-1.1986.0097.
- [36] ... H. G.-P. of the I. C. on and undefined 2003, "From a different perspective: principles, practice and potential of bistatic radar," *ieeexplore.ieee.org*, Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/1278701/>.
- [37] Y. Li, C. Rizos, E. Donskoi, J. Homer, and B. Mojarrabi, "3D Multi-static SAR System for Terrain Imaging Based on Indirect GPS Signals," 2002. Accessed: Jan. 15, 2021. [Online]. Available:
<http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.469.5668&rep=rep1&type=>

pdf.

- [38] B. Mojarrabi, J. Homer, ... K. K.-... and R. S., and undefined 2002, "Power budget study for passive target detection and imaging using secondary applications of GPS signals in bistatic radar systems," *ieeexplore.ieee.org*, Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/1025069/>.
- [39] M. M.-N.-E. journal and undefined 1993, "A passive reflectometry and interferometry system (PARIS): Application to ocean altimetry," *spotlight.unavco.org*, Accessed: Jan. 15, 2021. [Online]. Available: https://spotlight.unavco.org/how-gps-works/gps-basics/reflected-gps-signals_files/Martin-Neira-PARIS-1993.pdf.
- [40] J. L. Garrison, A. Komjathy, V. U. Zavorotny, and S. J. Katzberg, "Wind Speed Measurement Using Forward Scattered GPS Signals," 2002. Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/981349/>.
- [41] A. Komjathy, J. Maslanik, V. Zavorotny, and P. Axelrad, "Towards GPS surface reflection remote sensing of sea ice conditions," 2000, Accessed: Jan. 15, 2021. [Online]. Available: <https://ntrs.nasa.gov/search.jsp?R=20010124078>.
- [42] C. Zuffada, ... V. Z. P. and R. the F., and undefined 2001, "Coherence time and statistical properties of the GPS signal scattered off the ocean surface and their impact on the accuracy of remote sensing of sea surface," *ieeexplore.ieee.org*, Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/978344/>.
- [43] "(No Title)." <http://www2.unb.ca/gge/Resources/gpsworld.october07.pdf> (accessed Jan. 15, 2021).
- [44] "Student reflected GPS experiment (SuRGE)," *ieeexplore.ieee.org*, Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/976897/>.

- [45] J. G.-I. 2001. S. the P. and and undefined 2001, "Simulation of wind speed retrievals from an orbiting bistatic GPS receiver," *ieeexplore.ieee.org*, Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/976896/>.
- [46] D. Masters, S. Katzberg, P. A.-I. 2003. 2003 IEEE, and undefined 2003, "Airborne GPS bistatic radar soil moisture measurements during SMEX02," *ieeexplore.ieee.org*, Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/1293956/>.
- [47] V. Zavorotny *et al.*, "Seasonal Polarimetric Measurements of Soil Moisture Using Tower-Based GPS Bistatic Radar," 2003. Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/1293916/>.
- [48] X. Liao, "THE UNIVERSITY O F CALGARY CARRIER PHASE BASED IONOSPHERE RECOVERY OVER A REGIONAL AREA GPS NETWORK," 2001. Accessed: Jan. 15, 2021. [Online]. Available: <https://prism.ucalgary.ca/bitstream/handle/1880/40671/65000Liao.pdf?sequence=1>.
- [49] S. J. Katzberg and J. L. Garrison, "Utilizing GPS To Determine Ionospheric Delay Over the Ocean," 1996. Accessed: Jan. 15, 2021. [Online]. Available: <https://ntrs.nasa.gov/search.jsp?R=19970005019>.
- [50] H. D. Griffiths, "Bistatic and Multistatic Radar." Accessed: Jan. 15, 2021. [Online]. Available: http://www.ittc.ku.edu/~callen/energy_harvesting/Griffiths2003pp9.pdf.
- [51] H. Griffiths, C. Baker, H. Ghaleb, R. R.-E. Letters, and undefined 2003, "Measurement and analysis of ambiguity functions of off-air signals for passive coherent location," *IET*, Accessed: Jan. 15, 2021. [Online]. Available: https://digital-library.theiet.org/content/journals/10.1049/el_20030646.

- [52] D. Tan, H. Sun, Y. L.-I. I. R. Conference, and undefined 2005, "Sea and air moving target measurements using a GSM based passive radar," *ieeexplore.ieee.org*, Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/1435932/>.
- [53] E. Glennon, A. Dempster, C. R.- Positioning, and undefined 2006, "Feasibility of air target detection using GPS as a bistatic radar," *scirp.org*, Accessed: Jan. 15, 2021. [Online]. Available: <https://www.scirp.org/html/343.html>.
- [54] E. Donskoi, Y. Li, C. Rizos, J. Homer, and B. Mojarrabi, "Terrain imaging using a SAR system based on reflected GPS signals," *Artic. Wuhan Univ. J. Nat. Sci.*, 2003, doi: 10.1007/BF02899831.
- [55] M. Soumekh, *Synthetic aperture radar signal processing*. 1999.
- [56] G. C.-A. and P. S. International and undefined 1990, "On the use of the gradient to determine bistatic SAR resolution," *ece.ubc.ca*, Accessed: Jan. 15, 2021. [Online]. Available: <http://www.ece.ubc.ca/~yewn/papers/Cardillo90.pdf>.
- [57] T. Zeng and M. Cherniakov, "Generalized approach to resolution analysis in BSAR Passive Bistatic Radar View project FSR ground clutter View project Generalized Approach to Resolution Analysis in BSAR," *ieeexplore.ieee.org*, 2005, doi: 10.1109/TAES.2005.1468741.
- [58] M. Cherniakov, D. Nezlin, K. K.-I. Proceedings-Radar, and undefined 2002, "Air target detection via bistatic radar based on LEOS communication signals," *ieeexplore.ieee.org*, Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/993570/>.
- [59] F. ; Santi, M. ; Bucciarelli, D. ; Pastina, M. ; Antoniou, and M. Cherniakov, "Spatial

- resolution improvement in GNSS-based SAR using multi-static acquisitions and feature extraction,” *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, p. 6217, 7515, doi: 10.1109/TGRS.2016.2583784.
- [60] K. Bhataria, B. S.-2018 S. I. Conference, and undefined 2018, “A Review of Image Fusion Techniques,” *ieeexplore.ieee.org*, Accessed: Jan. 15, 2021. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/8487686/>.
- [61] G. Simone, A. Farina, F. C. Morabito, S. B. Serpico, and L. Bruzzone, “IMAGE FUSION TECHNIQUES FOR REMOTE SENSING APPLICATIONS,” 2002. Accessed: Jan. 15, 2021. [Online]. Available: <http://www.dit.unitn.it>.
- [62] K. Amolins, Y. Zhang, and P. Dare, “Wavelet based image fusion techniques - An introduction, review and comparison,” *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 62, no. 4. pp. 249–263, Sep. 2007, doi: 10.1016/j.isprsjprs.2007.05.009.